SADINhub Data Dictionary

- 1. Architecture Overview
- 2. Silver Staging: Core Dimensions
  - `silver.dim_energy_meter_staging`
  - `silver.dim_meter_cb_rating`
  - `silver.energy_meter_mapping_audit_staging`
  - `silver.dim_grid_billing_account_staging`
- 3. Confirmed Zone Mapping Corrections
- 4. Reporting Cadence
- 5. `silver.fact_energy_readings_staging`
  - 5.1 Confirmed unpopulated measurements (facility-wide)
- 6. Reference: Bronze source structure
- 7. Classification System
  - 7.1 Severity tiers
  - 7.2 What is classified, and what is not
  - 7.3 Current: authoritative production logic
  - 7.4 Function reference
- 8. `silver.fact_energy_readings_flagged`
- 9. `silver.fact_energy_readings_snapshot`
- 10. `silver.fact_energy_consumption_validated` (Consumption Calculation)
  - 10.1 Implemented validation rules
  - 10.2 Key reporting columns
  - 10.3 Validated volume and reconciliation
  - 10.4 Gold / Power BI integration
- 11. Gold Reporting Objects
  - 11.1 `gold.energy_raw`
  - 11.2 `gold.energy_consumption_validated`
  - 11.3 `gold.energy_reading_gaps`
- 12. Billing Objects (Not Yet Integrated)
- 13. Out of Scope
- 14. Governance & Documentation Status
- 15. Technical Object Registry
- 16. Formal Data-Quality Rule Registry
  - 16.1 Status-ranking utility
- 17. Data Lineage Registry
- 18. Physical Keys, Constraints & Indexes
  - 18.1 Confirmed indexes
- 19. Complete Column-Level Catalog
  - 19.1 `bronze.energy_meters`
  - 19.2 `bronze.telemetries`
  - 19.3 `bronze.telemetries_dictionaries`
  - 19.4 `gold.billing_account`
  - 19.5 `gold.date`
  - 19.6 `gold.energy_bill`
  - 19.7 `gold.energy_consumption_validated`
  - 19.8 `gold.energy_raw`

    - 19.9 `gold.energy_reading_gaps`
    - 19.10 `gold.facility`
    - 19.11 `gold.mapping_audit`
    - 19.12 `gold.meter`
    - 19.13 `gold.quality_summary`
    - 19.14 `gold.zone`
    - 19.15 `silver.dim_energy_meter_staging`
    - 19.16 `silver.dim_grid_billing_account_staging`
    - 19.17 `silver.dim_meter_cb_rating`
    - 19.18 `silver.energy_meter_mapping_audit_staging`
    - 19.19 `silver.fact_energy_consumption_validated`
    - 19.20 `silver.fact_energy_readings_flagged`
    - 19.21 `silver.fact_energy_readings_snapshot`
    - 19.22 `silver.fact_energy_readings_staging`
- 20. Metadata Evidence & Change Control
    - 20.1 Evidence used for this edition
    - 20.2 Important implementation confirmations from the live metadata
    - 20.3 Change-control principle

SADINhub Data Dictionary Energy Intelligence & Predictive Analytics Platform

Scope: Electrical energy data only (water-related tables are explicitly out of scope for this project).

Last updated 2026-08-24: live PostgreSQL metadata extracted and incorporated: production current-classifier overload verified, `overall_status` inputs corrected, object types/keys/indexes confirmed, and formal object/DQ/lineage/column catalogs added. Governance fields are documented only where established; no DAMA-DMBOK or ISO/IEC 11179 compliance is claimed.

1. Architecture Overview

Layer	Purpose	Location
Bronze	Raw source data, restored unchanged from `ssbms.public`	schema `bronze`
Silver	Technical normalization, canonical mapping, quality classification	schema `silver`
Gold	Reporting/BI-facing objects consumed by Power BI	schema `gold`

Source volume: ~107 Bronze tables, ~109M raw telemetry rows, 32 electrical telemetry keys, 31 canonical energy meters, 1 facility.

Core principle: raw grain, end to end. The pipeline serves data at its native reading grain (~5 minutes) from Bronze through to Gold. No pre-aggregation is performed in the database; any summarization by hour, day, week, or month happens in the BI layer. This keeps one source of truth, preserves full precision for pinpoint investigation, and means a reported value can always be traced back to a specific meter at a specific timestamp.

Bronze is never modified. Silver landing views perform technical normalization only; no trimming, dedup, interpolation, or permanent unit conversion. Canonical business-rule corrections (zone mapping fixes) happen only in Silver staging views.

Gold contents:

- `gold.energy_raw`: the raw fact (measured values + quality status), one row per meter per timestamp
- `gold.energy_consumption_validated`: validated interval-level `kWh` consumption derived from the cumulative register; see §10
- `gold.energy_reading_gaps`: reporting-gap evidence derived from consecutive same-meter reading timestamps; see §11
- `gold.meter`, `gold.zone`, `gold.facility`, `gold.billing_account`, `gold.date`: dimensions
- `gold.energy_bill`, `gold.mapping_audit`, `gold.quality_summary`: supporting objects

2. Silver Staging: Core Dimensions

`silver.dim_energy_meter_staging`

One canonical row per reporting device/meter. Joins meter and device metadata, assigns facility/zone/billing relationships, and resolves confirmed mapping exceptions for Zones 03, 13, and 20.

Key columns: `energy_meter_key` (uuid, logical unique key; this is a view, so PostgreSQL does not enforce a primary key), `source_energy_meter_id`, `facility_key`, `zone_key`, `grid_billing_account_key`, `parent_energy_meter_source_id`, `meter_label`, `meter_serial_number`, `phase_count`, `maximum_kwh`, `energy_meter_type_name`, `is_meter_active`, `device_name`, `device_label`, `device_type_name`, `device_category_name`, `is_device_active`, `is_device_online`, `mapping_status`.

`mapping_status` values: `TEAM_CONFIRMED_NAME_CORRECTION` (Zone 03), `TEAM_CONFIRMED_EXCEPTION_ONE_PANEL_TWO_METERS` (Zone 13), `VALID`, `REVIEW_NAME_MISMATCH`, `REVIEW_DUPLICATE_SOURCE_MAPPING_CANONICAL_NAME_MATCH`, `REVIEW_DUPLICATE_SOURCE_MAPPING_CANONICAL_SELECTED`.

Note on Zone 20: the live view excludes one confirmed duplicate mapping via a direct `WHERE NOT (device_id = '9574e049-9fb7-46db-9f8c-b3da0666aea0' AND meter_label = 'ZN.20-PB.2_child__1_ch3pm')`. Retained Zone 20 rows report `mapping_status = 'VALID'`.

Note on `energy_meter_key`: this value is identical to Bronze's `device_id`, verified by joining `bronze.telemetries.device_id` directly to `energy_meter_key` across all 31 meters with exact packet-count matches. No ID remapping occurs between Bronze and Silver.

`silver.dim_meter_cb_rating`

Per-meter circuit breaker maximum current rating. One row per canonical meter (31 rows).

Column	Type	Description
`energy_meter_key`	uuid	Primary key
`cb_max_rating_a`	double precision	That meter's real breaker rating, in amps
`updated_at`	timestamp	Last modification

Values range from 80A to 250A depending on the panel. This table exists because current classification cannot use a single facility-wide threshold; see §7.3.

No foreign key to `dim_energy_meter_staging`: that object is a view, and PostgreSQL foreign keys can only reference tables. Referential correctness is enforced at load time by joining on `meter_label`.

`silver.energy_meter_mapping_audit_staging`

Audit trail preserving every candidate source mapping, including excluded ones, with `business_rule_code`, `business_priority`, `canonical_mapping_rank`, `mapping_decision` (`KEEP` / `EXCLUDE`), and `mapping_status`.

`silver.dim_grid_billing_account_staging`

Billing account dimension. Exposes `breaker_capacity`, `minimal_breaker_capacity`, `maximum_breaker_capacity`, `service_class_fee`, `multiplication_factor`, and configured billing period fields.

3. Confirmed Zone Mapping Corrections

Zone	Issue	Resolution	Status
03	Source device name `ZN.3-PB_child__1_ch3pm` omitted `.1`, mismatching meter label `ZN.3-PB.1_child__1_ch3pm`	Canonical device name corrected to `ZN.3-PB.1_child__1_ch3pm`	`TEAM_CONFIRMED_NAME_CORRECTION`
13	Device `ZN.13-PB.1_child__2_ch3pm` legitimately maps to meter `ZN.13-PB.2_child__1_ch3pm` — one panel serving two meters	Kept as deliberate exception	`TEAM_CONFIRMED_EXCEPTION_ONE_PANEL_TWO_METERS`
20	Duplicate/crossed mapping: device `ZN.20-PB.1_child__1_ch3pm` incorrectly also mapped to meter `ZN.20-PB.2_child__1_ch3pm`	Duplicate excluded; genuine PB1→PB1 and PB2→PB2 mappings retained	`VALID` (retained rows)

All corrections occur in Silver staging only; Bronze and Silver landing views are unaffected.

Note: Zone 13 also appears separately in §10 for an unrelated reason: a kWh data-quality exclusion, not a mapping issue. The two are independent facts about Zone 13.

4. Reporting Cadence

- **Nominal/contractual assumption:** 300 seconds, 288 readings/day
- **Most meters:** ~301.5 seconds, ~286–287 readings/day
- **Zone 13 meters:** ~321.17–321.19 seconds, ~269 readings/day. This is a distinct, stable cadence, intentionally not normalized, since it reflects genuine device behavior

Reading timestamps carry real sub-second precision (e.g. `00:04:31.016`) and are never rounded or bucketed. Gaps in reporting appear as genuinely absent rows.

5. `silver.fact_energy_readings_staging`

The central analytical fact. **Grain:** one canonical meter × one telemetry timestamp. Pivots the long-format telemetry stream (`silver.telemetries_landing`, joined to `telemetries_dictionaries_landing` and `dim_energy_meter_staging`) into named columns.

32 telemetry keys across 9 classification groups:

Category	Keys	Notes
Voltage (Line-to-Line)	`V1,V2,V3,Va` → `voltage_line_1_v...average_v`	~380–420V typical
Voltage (Line-to-Neutral)	`Vn1,Vn2,Vn3,Vna` → `voltage_neutral_1_v...average_v`	~215–245V typical
Current	`Amp1,Amp2,Amp3,Ampa` → `current_phase_1_a...average_a`	
Frequency	`Hz` → `frequency_hz`	
Power Factor	`PF1,PF2,PF3,PFa` → `power_factor_phase_1...average`	
Active Power	`kW1,kW2,kW3,TkW` → `active_power_phase_1_kw...total_kw`	Instantaneous, not cumulative
Apparent Power	`kVA1,kVA2,kVA3,TkVA` → `apparent_power_phase_1_kva...total_kva`	
Reactive Power	`kVAr1,kVAr2,kVAr3,TkVAr` → `reactive_power_phase_1_kvar...total_kvar`	Any sign allowed (leading/lagging)
Accumulated Energy	`kWh, kVAh, kVArh` → `*_energy_cumulative_*`	Cumulative counters — never differenced in this view (a downstream table does perform differencing — see §10)

Also includes `date_key`, `time_key`, `reading_timestamp_utc`, `reading_timestamp_saudi` / `reading_date_saudi` (Asia/Riyadh, Friday–Saturday weekend), `source_measurement_row_count`, `populated_measurement_count`, `duplicate_measurement_row_count`, and derived phase-imbalance diagnostics.

5.1 Confirmed unpopulated measurements (facility-wide)

Eight of the 32 keys were **not observed for any of the 31 meters in the validated Bronze sample/window**: `Va`, `Vna`, `Ampa`, and the entire apparent-power/apparent-energy family (`kVA1`, `kVA2`, `kVA3`, `TkVA`, `kVAh`).

Verified at the Bronze level: a distinct-key count per meter in the validated sample/window returned exactly 24 keys for every meter, with none reporting more. This is consistent with fleet-wide meter hardware/firmware behavior rather than a coverage problem in specific meters. The downstream columns are `NULL` throughout the analytical layer and are excluded from classification (§7.2). The check has not yet been extended across the full 19-month Bronze history, so this wording deliberately does not claim that the keys were never transmitted at any point.

6. Reference: Bronze source structure

`bronze.telemetries` is long-format: one row per individual measurement, not per reading:

Column	Type	Description
`device_id`	uuid	Matches `energy_meter_key` (§2)
`key_id`	bigint	Which measurement this row holds
`timestamp`	bigint	Epoch milliseconds
`bool`, `dbl`, `json`, `long`, `str`	various	Typed value columns; electrical measurements use `dbl`

One “reading packet” is therefore a unique `(device_id, timestamp)` pair spanning ~24 rows. Silver’s pivot reshapes these into one wide row per packet.

7. Classification System

7.1 Severity tiers

Five tiers, in increasing severity:

`Acceptable` < `Review Required` < `Problematic` < `Critical` < `Invalid`

Evaluation order: first match wins:

`Invalid` → `Critical` → `Review Required` → `Problematic` → `Acceptable`

Current also has two diagnostic outcomes outside this five-tier ladder: `Major Issue` and `Major Problem` (§7.3). They identify current/active-power consistency mismatches and do not elevate `overall_status`.

This order matters: several tiers have mathematically overlapping conditions, and the order resolves which label applies. `Invalid` is checked first so a corrupt or missing reading is never mislabeled as merely out-of-range. `Review Required` is checked before `Problematic` so the narrower, less severe band is carved out of the wider one rather than being swallowed by it.

`silver.status_rank(status)` currently recognizes `Acceptable`=1, `Review Required`=2, `Anomaly`=3, `Problematic`=4, `Critical`=5, and `Invalid`=6 (0 for unrecognized text). The current v2 classifiers documented below do not emit `Anomaly`, but the ranking utility retains support for that label. `silver.worst_status(VARIADIC statuses)` filters `NULL` inputs before ranking and returns the highest-ranked remaining status; if every input is `NULL`, it returns `NULL`.

7.2 What is classified, and what is not

Classified: Voltage (Line-to-Line and Line-to-Neutral, phases 1–3), Current (phases 1–3), Frequency, Power Factor (phases 1–3 and average), Accumulated active energy (`kWh`).

Not classified:

Category	Reason
Active Power (`kW1-3`, `TkW`)	Project decision — no classification required
Apparent Power (`kVA1-3`, `TkVA`)	Project decision; also never populated (§5.1)
	Project decision — no alarms set, both leading and lagging PF are valid

Category	Reason
Reactive Power (`kVAr1-3`, `TkVAr`)	
Accumulated `kVAh`, `kVArh`	Belong to the unclassified apparent/reactive families
`Va`, `Vna`, `Ampa`	Never populated by any meter (§5.1) — classifying an always-`NULL` value would make `overall_status` permanently `Invalid` for reasons unrelated to any real condition

Raw measured values for all of the above remain fully available throughout the pipeline. Only their *status evaluation* is omitted.

7.3 Current: authoritative production logic

Current is the only classified measurement whose threshold depends on the individual meter's circuit-breaker rating. `silver.dim_meter_cb_rating` stores one `cb_max_rating_a` value per canonical meter.

Authoritative production call: the live `silver.fact_energy_readings_flagged` view calls the three-argument overload:

```
silver.classify_current_v2(current_value, cb_max_rating_a, same_phase_active_power_kw)
```

A separate two-argument overload, `classify_current_v2(value, cb_max)`, also exists in the database, but the current flagged analytical view does **not** call it. The three-argument overload is therefore the production rule for `status_amp1`, `status_amp2`, and `status_amp3`.

Given breaker rating `CB`:

```
margin = 0.15 × CB
third  = margin ÷ 3
```

The production function evaluates in this exact order:

Order	Condition	Result
1	current or breaker rating is `NULL`	`Invalid`
2	current `< 0`	`Invalid`
3	current `> CB`	`Critical`
4	current `> CB − third`	`Problematic`
5	current `> CB − margin`	`Review Required`
6	current `< margin` and current = 0 and same-phase kW = 0	`Acceptable`
7	current `< margin` and current > 0 and same-phase kW = 0	`Major Problem`
8	current `< margin` and current = 0 and same-phase kW <> 0	`Major Issue`
9	remaining lower-fringe cases	`Acceptable`
10	remaining middle-range cases	`Acceptable`

Important scope of the current/power consistency check: `Major Issue` / `Major Problem` are evaluated only inside the lower fringe (`current < 15% of CB`). They are not global comparisons applied at every current magnitude.

Zero semantics: the production function uses literal numeric comparisons (`= 0`, `<> 0`, `> 0`). No epsilon/tolerance band is applied. If `phase_kw` is `NULL`, none of the equality/inequality branches against zero evaluate true, so the lower-fringe fallback is `Acceptable`.

Boundary behavior from the live function: - exactly `CB` → `Problematic` (only `> CB` is `Critical`) - exactly `CB − third` → `Review Required` (the `Problematic` test is strict `>`) - exactly `CB − margin` → `Acceptable` (the `Review Required` test is strict `>`) - exactly `margin` → `Acceptable` - current `< 0` → `Invalid`, not `Critical`

The validated facility results show breaker-proximity statuses only on Phase 1 of `ZN.1-PB.2` (179 `Review Required`, 26 `Problematic`, 9 `Critical`). The much larger current finding set comes from `Major Issue` / `Major Problem` lower-fringe consistency outcomes.

7.4 Function reference

All functions live in schema `silver`.

Function	Signature	Bands
`classify_voltage_line_v2`	`(value)`	Min/Max 360–440V · Acceptable 380–420V · Review Required 371.2–380 and 420–428.8 (2% of Max either side) · Problematic 360–371.2 and 428.8–440 · Critical outside 360–440 · Invalid on `NULL`
`classify_voltage_neutral_v2`	`(value)`	Min/Max 207–253V · Acceptable 215–245V · Review Required 209.94–215 and 245–250.06 (2% of Max) · Problematic 207–209.94 and 250.06–253 · Critical outside 207–253 · Invalid on `NULL`
`classify_current_v2`	`(value, cb_max)`	Two-argument overload present in the live database; applies breaker-relative low/high bands but is **not called by the current `fact_energy_readings_flagged` view**.
`classify_current_v2`	`(value, cb_max, phase_kw)`	**Production overload used by `fact_energy_readings_flagged`**. Applies upper breaker-margin rules plus lower-fringe same-phase current/kW consistency logic; see §7.3.
`classify_frequency_v2`	`(value)`	Min/Max 59–61Hz · Acceptable 59.3–60.5Hz · Problematic 59–59.3 and 60.5–61 · Critical outside 59–61 · Invalid on `NULL`. No Review Required tier for this metric.
`classify_power_factor_v2`	`(value)`	Min/Max 0.6–1 · Acceptable 0.8–1 · Review Required 0.78–0.8 · Problematic 0.6–0.78 · Critical `< 0.6` or `> 1` · Invalid on `NULL`. Acceptable Max equals Max, so there is no upper Review Required or Problematic band.
`classify_accumu-`	`(value, prev_`	`kWh` only. `Invalid` if the reading is lower than that meter's previous reading — a cumulative counter must never decrease. `Acceptable` if it increased or stayed flat, including the first-ever reading for a meter (nothing to compare against).

Function	Signature	Bands
`lated_energy_v2`	`value)`	

Accumulated energy: how the comparison works. `classify_accumulated_energy_v2` is the only classifier requiring a previous value. `silver.fact_energy_readings_flagged` supplies it via `LAG(active_energy_cumulative_kwh) OVER (PARTITION BY energy_meter_key ORDER BY reading_timestamp_utc)`. The `PARTITION BY` is essential: each meter is compared only against its own prior reading, never against a different meter's, regardless of how rows are physically ordered.

Each meter increments its `kWh` counter by its own fixed step (e.g. 0.4 on one meter, 0.5 on another), and the step size varies across the fleet. Because per-meter step values are not yet catalogued, the current check validates direction only; the counter must not go backward. A future refinement could add step-size validation (flagging an increment that is neither zero nor that meter's expected step) once those values are collected.

8. `silver.fact_energy_readings_flagged`

Row-level status classification view at the raw reading grain.

Columns: identity fields (`energy_meter_key`, `source_energy_meter_id`, `facility_key`, `zone_key`, `grid_billing_account_key`, `date_key`, `time_key`, `source_timestamp_ms`, `reading_timestamp_utc`, `reading_timestamp_saudi`, `reading_date_saudi`), then:

`status_vn1`, `status_vn2`, `status_vn3`, `status_v1`, `status_v2`, `status_v3`, `status_amp1`, `status_amp2`, `status_amp3`, `status_hz`, `status_pf1`, `status_pf2`, `status_pf3`, `status_pfa`, `status_kwh`, and `overall_status`.

This view is **status-only**; it does not carry the underlying measured values. It joins `silver.dim_meter_cb_rating` to obtain each meter's breaker rating for current classification (§7.3), evaluates same-phase current/active-power consistency for `status_amp1` / `status_amp2` / `status_amp3`, and computes `LAG()` for accumulated energy's decrement check (§7.4).

A note on `overall_status`: the live view calls `worst_status()` across **12 inputs only**: the six voltage statuses, frequency, PF1/PF2/PF3/PFa, and kWh. `status_amp1`, `status_amp2`, and `status_amp3` are excluded entirely from the `overall_status` array. Therefore **no current status**, including `Major Issue`, `Major Problem`, `Review Required`, `Problematic`, `Critical`, or `Invalid`, can change `overall_status`. Current must be analyzed through its individual phase status columns.

9. `silver.fact_energy_readings_snapshot`

A stored table (not a view) joining measured values to their quality status; the analytical table Gold serves.

Built from: `silver.fact_energy_readings_staging` (measured values) `JOIN` `silver.fact_energy_readings_flagged` (status) on `energy_meter_key` + `reading_timestamp_utc`.

Primary key: `(energy_meter_key, reading_timestamp_utc)`

Why a stored table rather than a view: `fact_energy_readings_flagged` recomputes the full Bronze pivot, classification, and window functions on every query. Materializing the result once means Power BI reads a plain indexed table instead of triggering that recomputation on every refresh.

Refresh method: full rebuild (`DROP TABLE ... CASCADE` then reload), not incremental. Accumulated energy's decrement check depends on each reading's true predecessor; an incremental filter would compare the first new reading against the wrong prior value at the boundary. A full rebuild avoids that risk entirely, at the cost of reprocessing the full history each time (~1–1.5 hours at current volume).

Important: the `CASCADE` also drops `gold.energy_raw`, which must be explicitly recreated after every rebuild:

```
CREATE OR REPLACE VIEW gold.energy_raw AS
SELECT * FROM silver.fact_energy_readings_snapshot;
```

Current volume: 4,535,918 rows, cross-validated by five independent methods: Bronze distinct-packet counts per meter (all 31 matching exactly), direct row counts against both source views, per-meter reading counts, status-distribution totals, and duplicate/null-identity checks (zero of each).

Consumption (differenced `kWh`) is calculated downstream of this table in the implemented and validated `silver.fact_energy_consumption_validated` table; see §10.

10. `silver.fact_energy_consumption_validated` (Consumption Calculation)

Status: implemented, validated, exposed through Gold, and used by Power BI.

Purpose: `classify_accumulated_energy_v2` (§7.4) checks only whether the cumulative `kWh` counter decreased. `silver.fact_energy_consumption_validated` converts the cumulative register into interval-level consumption while preserving the raw source values in `silver.fact_energy_readings_snapshot`.

Source: `silver.fact_energy_readings_snapshot`, joined to `silver.dim_energy_meter_staging` for the canonical `meter_label`.

10.1 Implemented validation rules

The current implementation applies these rules in sequence:

- **Zone 13 exclusion:** `ZN.13-PB.1` and `ZN.13-PB.2` are excluded from validated consumption. This is a kWh data-quality/business-rule exclusion and is independent of the Zone 13 mapping exception in §3.
- **`ZN.1-PB.1` startup exclusion:** readings before `2024-12-10 17:16:15.655 UTC` are ignored; the approved valid baseline begins at `46.5 kWh`.
- **`ZN.4-PB.1` startup exclusion:** readings before `2024-12-10 21:46:39.010 UTC` are ignored; the approved valid baseline begins at `150.4 kWh`.

- **Isolated low bad reading:** when a reading drops below the prior value and the following reading returns to or above the prior level, the isolated low point is ignored and consumption bridges between the surrounding valid readings.
- **Isolated high spike:** when a reading spikes above both its prior and following values and the following value recovers to or above the prior level, the spike is ignored and consumption bridges between the surrounding valid readings.
- **Normal increase:** validated consumption is the current valid cumulative `kWh` minus the previous valid cumulative `kWh`.
- **Flat counter:** contributes `0 kWh`.
- **Remaining decrement after validation:** not counted as validated consumption.
- **Long gap:** a valid-to-valid interval of at least 15 minutes is labeled `LONG_GAP`. The cumulative increase across that gap is intentionally **included** in validated consumption under the approved business rule.

Rule precedence: meter-level exclusions and startup cutoffs are applied first; isolated low/high bad points are removed before the valid-reading sequence is differenced; the validated delta is then evaluated. If the remaining valid-to-valid delta is negative, it is not counted as consumption. Only a non-negative valid interval is subsequently labeled `LONG_GAP` when its elapsed time is at least 15 minutes; otherwise it is `NORMAL_INTERVAL`. Therefore an unresolved decrement takes precedence over the long-gap label for consumption purposes.

Long gaps therefore affect **time attribution**, not the validity of the total: the accumulated increase is assigned to the first valid reading after telemetry resumes.

10.2 Key reporting columns

Key columns used by the reporting layer include:

- `energy_meter_key`
- `meter_label`
- `reading_timestamp_utc`
- `reading_date_saudi`
- `active_energy_cumulative_kwh`
- `validated_kwh_consumption`
- `interval_quality`

Observed `interval_quality` values in the validated dataset are `FIRST_VALID_READING`, `IGNORED_READING`, `LONG_GAP`, and `NORMAL_INTERVAL`.

10.3 Validated volume and reconciliation

Metric	Validated result
Rows	4,260,849
Included meters	29
Total validated consumption	2,020,697.5 kWh
`FIRST_VALID_READING` rows	29
`IGNORED_READING` rows	28
`LONG_GAP` rows	875

Metric	Validated result
`LONG_GAP` consumption	354,242.5 kWh
`NORMAL_INTERVAL` rows	4,259,917
`NORMAL_INTERVAL` consumption	1,666,455.0 kWh

All 29 included meters were reconciled individually against the approved cumulative-register baseline and final register value. Every meter returned a reconciliation difference of exactly `0.000000`.

Validation result: no residual reset-induced overcount or undercount was observed in the validated dataset. This reconciliation is the empirical check that the implemented isolated-point logic is sufficient for the 29 included meters in the assessed history; a more elaborate high-water-mark/reset model is therefore not required for the current validated dataset.

The Silver table and `gold.energy_consumption_validated` were also cross-checked directly and match exactly on row count, meter count, and total validated consumption. The underlying PostgreSQL double-precision sum is `2,020,697.4999726608 kWh`; reporting documentation rounds this to `2,020,697.5 kWh` to avoid implying false measurement precision.

10.4 Gold / Power BI integration

`gold.energy_consumption_validated` exposes this table to Power BI.

Power BI uses:

```
 Total Validated Consumption kWh =
SUM('gold energy_consumption_validated'[validated_kwh_consumption])
```

The consumption fact has an active many-to-one relationship to `gold.meter` on `energy_meter_key`. Direct relationships from the consumption fact to zone, facility, and billing account remain inactive to avoid ambiguous duplicate filter paths, and there is no direct fact-to-fact relationship to `gold.energy_raw`.

Monthly reporting uses a Power BI `Reading Month UTC` calculated column derived from `reading_timestamp_utc`. Weekday/weekend reporting uses `reading_date_saudi`, with Friday and Saturday classified as weekend.

11. Gold Reporting Objects

11.1 `gold.energy_raw`

The primary raw-reading Gold fact. A plain passthrough view over `silver.fact_energy_readings_snapshot`; no logic, no filtering, no aggregation.

Power BI model requirements:

1. **`reading_date` calculated column:** `reading_timestamp_utc` carries a full timestamp with sub-second precision, while `gold.date[calendar_date]` holds date-only values. These cannot relate directly, so a date-only column is derived:

```
reading_date = DATE(YEAR('gold energy_raw'[reading_timestamp_utc]), MONTH('gold energy_raw'[reading_timestamp_utc]), DAY('gold energy_raw'[reading_timestamp_utc]))
```

2. **`meter_label` calculated column:** surfaces the readable meter name (e.g. `ZN.16-PB_child__1_ch3pm`) alongside raw readings:

```
meter_label = RELATED('gold meter'[meter_label])
```

3. **Two active relationships:** `energy_meter_key` → `gold.meter` and `reading_date` → `gold.date[calendar_date]`, both Many-to-one, Single cross-filter direction

4. **Relationships that must stay inactive:** `facility_key` → `gold.facility`, `zone_key` → `gold.zone`, `grid_billing_account_key` → `gold.billing_account`, `energy_meter_key` → `gold.quality_summary`. The meter-hub relationship already reaches zone, facility, and billing account through `gold.meter`'s own hierarchy; activating these direct links creates an ambiguous duplicate path that Power BI rejects. When activating the meter relationship, deactivate these first; Power BI enforces this ordering.

11.2 `gold.energy_consumption_validated`

Gold view over `silver.fact_energy_consumption_validated` (§10). This is the reporting source for validated energy-consumption totals, per-meter consumption, monthly consumption, and weekday/weekend consumption analysis.

Current validated volume: 4,260,849 rows across 29 included meters, totaling 2,020,697.5 kWh.

Power BI relationship: active many-to-one from `energy_meter_key` to `gold.meter[energy_meter_key]`, single direction. Direct zone/facility/billing relationships remain inactive to avoid ambiguous paths. There is no direct relationship to `gold.energy_raw`.

11.3 `gold.energy_reading_gaps`

Physical Gold reporting table for telemetry coverage gaps.

Definition: for each meter, consecutive readings are ordered by `reading_timestamp_utc`; an interval is recorded as a gap when the elapsed time is at least 900 seconds (15 minutes).

Key reporting fields include `meter_label`, the previous and current reading timestamps, `gap_minutes`, and `gap_days`.

Refresh / staleness behavior: PostgreSQL metadata confirms this object is a physical table, not a view. The catalog extract does not contain its build procedure, scheduler, or refresh ownership, so those operational details remain **not formally documented**. Because it stores derived gap rows rather than evaluating a live view, it must be refreshed when the underlying reading history changes if the gap analysis is to remain current.

The largest validated gap is `ZN.17-PB`: `2025-05-18 04:22:28.966 UTC` → `2025-12-05 12:55:30.761 UTC`, equal to 289,953.03 minutes / 201.36 days.

12. Billing Objects (Not Yet Integrated)

Present in the schema but not yet connected to consumption analysis, pending real billing data and criteria:

- `silver.grid_billing_accounts_landing`, `grid_billing_accounts_service_classes_landing`, `grid_billing_accounts_types_landing`
- `silver.tariff_categories_landing`: a candidate source not yet incorporated into the star schema
- `silver.energy_bills_tracking_landing`, `silver.fact_energy_bill_staging`: a source-aligned bill tracking/comparison fact (grain: one tracked bill per billing period per grid billing account), exposing billed versus system-estimated consumption/cost and variances

Not a tariff-calculation engine; no verified tier, time-of-use, VAT, or demand-charge formula exists yet.

13. Out of Scope

All water-related tables (`water_bill*`, `water_grid_*`, `water_tank*`, `water_meter_units_landing`) are excluded from this project's analysis per project scope.

14. Governance & Documentation Status

This dictionary is **informed by common enterprise metadata-management practices** (including concepts commonly associated with DAMA-DMBOK and ISO/IEC 11179), but it does **not** claim formal compliance or certification against either framework.

Only governance facts established in the project evidence are populated below. Organizational assignments or policies that have not been formally confirmed are deliberately shown as `TBD / not formally defined` rather than inferred.

Governance item	Current documented state
Data domain	Electrical Energy Telemetry
Project scope	SADINhub Energy Intelligence & Predictive Analytics Platform; water data excluded
System of record for raw telemetry	PostgreSQL Bronze layer / source-restored telemetry
Analytical platform	PostgreSQL 18 + Power BI
Downstream consumers	Power BI dashboard; analytics/assessment workflows
Data owner	TBD / not formally assigned in project evidence
Data steward	TBD / not formally assigned in project evidence
Technical owner	TBD / not formally assigned in project evidence
Refresh SLA	TBD / not formally defined
Retention policy	TBD / not formally defined
Security / sensitivity classification	TBD / not formally assessed
Formal approval status	Project technical documentation; formal enterprise approval not evidenced

Governance item	Current documented state
Technical validation status	Validated for the documented assessment scope
Metadata extraction date	2026-08-24
Catalog identity	`(schema_name, object_name, column_name)`

15. Technical Object Registry

Object type and size below come directly from the live PostgreSQL catalog at extraction time. View size is shown as `0 bytes` because the view itself stores no rows.

Schema	Object	Type	Catalog size	Comment / purpose
bronze	energy_meters	table	144 kB	
bronze	telemetries	table	12 GB	
bronze	telemetries_dictionaries	table	40 kB	
gold	billing_account	view	0 bytes	
gold	date	view	0 bytes	
gold	energy_bill	view	0 bytes	
gold	energy_consumption_validated	view	0 bytes	
gold	energy_raw	view	0 bytes	
gold	energy_reading_gaps	table	136 kB	
gold	facility	view	0 bytes	
gold	mapping_audit	view	0 bytes	
gold	meter	view	0 bytes	
gold	quality_summary	view	0 bytes	
gold	zone	view	0 bytes	
silver	dim_energy_meter_staging	view	0 bytes	Canonical energy-meter/device dimension. One row per reporting device; duplicate source mappings remain flagged.
silver	dim_grid_billing_account_staging	view	0 bytes	Grid billing-account dimension enriched with account type and service class.
silver	dim_meter_cb_rating	table	24 kB	
silver	energy_meter_mapping_audit_staging	view	0 bytes	
silver	fact_energy_consumption_validated	table	1221 MB	
silver	fact_energy_readings_flagged	view	0 bytes	

Schema	Object	Type	Catalog size	Comment / purpose
silver	fact_energy_readings_snapshot	table	2287 MB	
silver	fact_energy_readings_staging	view	0 bytes	Main project fact. One energy meter per telemetry timestamp with electrical measures pivoted into business-readable columns.

16. Formal Data-Quality Rule Registry

These rule IDs organize logic that already exists in the pipeline; they do not create new thresholds or change the live database behavior.

Rule ID	Rule name	Scope	Authoritative logic / definition	Primary outputs	Status
DQ-MAP-001	Canonical meter mapping	Meter/device mapping	Canonical ranking plus confirmed Zone 03/13/20 business rules.	Mapping/audit fields	Implemented
DQ-PACKET-001	Telemetry packet completeness diagnostics	Reading packet	Counts source rows, distinct populated keys, and duplicate measurement rows per meter×timestamp.	source_measurement_row_count, populated_measurement_count, duplicate_measurement_row_count	Implemented diagnostic
DQ-VL-001	Line voltage classification	V1/V2/V3	Critical <360 or >440; Acceptable 380–420; Review 371.2–<380 or >420–428.8; remaining in-range values Problematic; NULL Invalid.	status_v1/ status_v2/ status_v3	Implemented
DQ-VN-001	Line-to-neutral voltage classification	Vn1/Vn2/Vn3	Critical <207 or >253; Acceptable 215–245; Review 209.94–<215 or >245–250.06; remaining in-range values Problematic; NULL Invalid.	status_vn1/ status_vn2/ status_vn3	Implemented
DQ-FREQ-001	Frequency classification	Hz	Critical <59 or >61; Acceptable 59.3–60.5; remaining 59–61 values Problematic; NULL Invalid.	status_hz	Implemented
DQ-PF-001	Power-factor classification	PF1/PF2/PF3/PFa	Critical <0.6 or >1; Acceptable 0.8–1; Review 0.78–<0.8; remaining 0.6–	status_pf1/ status_pf2/ status_pf3/ status_pfa	Implemented

Rule ID	Rule name	Scope	Authoritative logic / definition	Primary outputs	Status
			<0.78 Problematic; NULL Invalid.		
DQ-CUR-001	Per-meter current + same-phase power consistency	Amp1/Amp2/Amp3 + kW1/kW2/kW3 + breaker rating	Production 3-argument classifier; exact order and strict boundaries documented in §7.3.	status_amp1/ status_amp2/ status_amp3	Implemented
DQ-KWH-001	Cumulative active-energy direction	kWh	Invalid when current cumulative kWh < previous same-meter kWh; first reading and non-decreasing values Acceptable.	status_kwh	Implemented
DQ-CONS-001	Validated interval consumption	Cumulative kWh	Approved exclusions, isolated-point cleaning, valid-to-valid differencing, decrement suppression, and long-gap inclusion.	validated_kwh_consumption, consumption_cleaning_status, interval_quality	Implemented + reconciled
DQ-GAP-001	Telemetry coverage gap	Reading timestamps	Physical gap rows for same-meter elapsed intervals ≥900 seconds (15 minutes).	gap_seconds/ minutes/hours/days	Implemented
DQ-IMB-CUR-001	Current imbalance diagnostic	Amp1/Amp2/Amp3	3-phase average/ spread/imbalance computed when all phases are present and absolute average current >0.1A.	current_* imbalance fields	Implemented diagnostic
DQ-IMB-VOLT-001	Voltage imbalance diagnostic	V1/V2/V3	3-phase average/ spread/imbalance computed when all phases are present and absolute average voltage >1.0V.	voltage_* imbalance fields	Implemented diagnostic

16.1 Status-ranking utility

The live `silver.status_rank()` function recognizes:

`Acceptable` (1) < `Review Required` (2) < `Anomaly` (3) < `Problematic` (4) < `Critical` (5) < `Invalid` (6).

`Anomaly` remains supported by the ranking utility even though the current v2 classifiers extracted on 2026-08-24 do not emit it. Unrecognized text ranks `0`.

`overall_status` is **not** a rollup of every `status_*` column. The current live view explicitly excludes `status_amp1`, `status_amp2`, and `status_amp3`.

17. Data Lineage Registry

Target object	Primary source(s)	Transformation / lineage
silver.dim_energy_meter_staging	Silver landing meter/device/unit/type/category views	Joins source metadata, applies Zone 03 name correction, Zone 13 confirmed exception, Zone 20 duplicate exclusion, and canonical ranking.
silver.energy_meter_mapping_audit_staging	Silver landing meter/device/unit views	Preserves all candidate mappings and records business rule, rank, KEEP/EXCLUDE decision, and mapping status.
silver.dim_grid_billing_account_staging	Silver billing-account/type/service-class landing views	Enriches billing account with type, service class, capacities, fee and billing-period configuration.
silver.fact_energy_readings_staging	silver.telemetries_landing + telemetries_dictionaries_landing + dim_energy_meter_staging	Filters the 32 electrical telemetry keys, converts epoch-ms to UTC/Saudi timestamps, pivots long telemetry to one meter×timestamp row, and derives imbalance diagnostics.
silver.fact_energy_readings_flagged	fact_energy_readings_staging + dim_meter_cb_rating	Applies voltage/current/frequency/PF/kWh classifiers. kWh uses meter-partitioned LAG. Current uses the 3-argument classifier overload.
silver.fact_energy_readings_snapshot	fact_energy_readings_staging + fact_energy_readings_flagged	Stored raw-grain analytical snapshot combining measurements and status columns.
silver.fact_energy_consumption_validated	fact_energy_readings_snapshot + dim_energy_meter_staging	Applies approved cumulative-kWh cleaning/differencing rules and interval-quality classification.
gold.energy_raw	silver.fact_energy_readings_snapshot	Pass-through reporting view; no additional SQL transformation.
gold.energy_consumption_validated	silver.fact_energy_consumption_validated	Pass-through reporting view exposing 15 BI-facing consumption fields.
gold.meter	silver.dim_energy_meter_staging	Pass-through reporting dimension.
gold.billing_account	silver.dim_grid_billing_account_staging	Pass-through reporting dimension.
gold.mapping_audit	silver.energy_meter_mapping_audit_staging	Pass-through reporting audit view.
gold.energy_reading_gaps	Consecutive meter readings from analytical reading history	Physical derived table containing intervals ≥15 minutes; exact build/refresh procedure is not captured by the supplied PostgreSQL catalog metadata.

18. Physical Keys, Constraints & Indexes

Views do not carry enforced PostgreSQL primary/foreign-key constraints. The following are the physical constraints confirmed in the extracted object scope; column nullability is recorded separately in §19.

Object	Constraint type	Constraint name	Definition
bronze.energy_meters	PRIMARY KEY	energy_meters_pkey	PRIMARY KEY (id)
bronze.energy_meters	CHECK		maximum_kwh > 0

Object	Constraint type	Constraint name	Definition
		energy_meters_maximum_kwh_check	
bronze.energy_meters	CHECK	energy_meters_phase_check	phase > 0
bronze.energy_meters	FOREIGN KEY	energy_meters_created_by_fkey	created_by → bronze.users(id)
bronze.energy_meters	FOREIGN KEY	energy_meters_device_id_fkey	device_id → bronze.device(id)
bronze.energy_meters	FOREIGN KEY	energy_meters_facility_id_fkey	facility_id → bronze.facilities(id)
bronze.energy_meters	FOREIGN KEY	energy_meters_grid_billing_account_id_fkey	grid_billing_account_id → bronze.grid_billing_accounts(id)
bronze.energy_meters	FOREIGN KEY	energy_meters_parent_id_fkey	parent_id → bronze.energy_meters(id)
bronze.energy_meters	FOREIGN KEY	energy_meters_type_id_fkey	type_id → bronze.energy_meters_types(id)
bronze.energy_meters	FOREIGN KEY	energy_meters_unit_id_fkey	unit_id → bronze.units(id)
bronze.telemetries	PRIMARY KEY	telemetries_pkey	PRIMARY KEY (device_id, key_id, “timestamp”)
bronze.telemetries_dictionaries	PRIMARY KEY	telemetries_dictionaries_pkey	PRIMARY KEY (id)
silver.dim_meter_cb_rating	PRIMARY KEY	dim_meter_cb_rating_pkey	PRIMARY KEY (energy_meter_key)
silver.fact_energy_readings_snapshot	PRIMARY KEY	pk_energy_readings_snapshot	PRIMARY KEY (energy_meter_key, reading_timestamp_utc)

18.1 Confirmed indexes

Object	Index name	Definition
bronze.energy_meters	energy_meters_pkey	UNIQUE btree(id)
bronze.energy_meters	energymeter_created_by	btree(created_by)
bronze.energy_meters	energymeter_device_id	btree(device_id)
bronze.energy_meters	energymeter_facility_id	btree(facility_id)
bronze.energy_meters	energymeter_grid_billing_account_id	btree(grid_billing_account_id)
bronze.energy_meters	energymeter_parent_id	btree(parent_id)
bronze.energy_meters	energymeter_type_id	btree(type_id)
bronze.energy_meters	energymeter_unit_id	btree(unit_id)
bronze.telemetries	telemetries_pkey	UNIQUE btree(device_id, key_id, “timestamp”)
bronze.telemetries_dictionaries	telemetries_dictionaries_pkey	UNIQUE btree(id)
bronze.telemetries_dictionaries	telemetrydictionaries_key	UNIQUE btree(key)
silver.dim_meter_cb_rating	dim_meter_cb_rating_pkey	UNIQUE btree(energy_meter_key)

Object	Index name	Definition
silver.fact_energy_consumption_validated	idx_consumption_date	btree(reading_date_saudi)
silver.fact_energy_consumption_validated	idx_consumption_meter_label	btree(meter_label)
silver.fact_energy_consumption_validated	idx_consumption_meter_time	btree(energy_meter_key, reading_timestamp_utc)
silver.fact_energy_readings_snapshot	pk_energy_readings_snapshot	UNIQUE btree(energy_meter_key, reading_timestamp_utc)

Consumption-table note: `silver.fact_energy_consumption_validated` has three confirmed non-unique indexes but no primary-key constraint in the extracted metadata. Its `(energy_meter_key, reading_timestamp_utc)` pair is indexed, but the database does not enforce uniqueness on that pair through a PK/UNIQUE constraint in the supplied catalog output.

19. Complete Column-Level Catalog

Catalog key: `(schema, object, column)`. The same column name may have different grain, nullability, constraints, or semantics in different objects, so column name alone is not treated as a unique identifier.

`Nullable` and PostgreSQL data type come directly from `information_schema.columns`. `Key` and `Indexed` are based on the extracted physical constraints/indexes in scope. Business definitions, units, lineage, and DQ mappings describe the already-documented SADINhub pipeline and do not override the physical database metadata.

19.1 `bronze.energy_meters`

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
1	id	uuid	NO	gen_random_uuid()	PK	Yes	—	Id.	Source-system Bronze field; no analytical transformation.	—
2	facility_id	uuid	NO	—	—	Yes	—	Facility id identifier.	Source-system Bronze field; no analytical transformation.	—
3	grid_billing_account_id	uuid	NO	—	—	Yes	—	Grid billing account id identifier.	Source-system Bronze field; no analytical trans-	—

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
									formation.	
4	device_id	uuid	NO	—	—	Yes	—	Device id identifier.	Source-system Bronze field; no analytical transformation.	—
5	phase	integer	NO	—	—	No/Not shown	—	Phase.	Source-system Bronze field; no analytical transformation.	—
6	parent_id	uuid	YES	—	—	Yes	—	Parent id identifier.	Source-system Bronze field; no analytical transformation.	—
7	maximum_kwh	integer	NO	—	—	No/Not shown	kWh	Maximum kwh.	Source-system Bronze field; no analytical transformation.	—
8	type_id	uuid	NO	—	—	Yes	—	Type id identifier.	Source-system Bronze field; no analytical transformation.	—
9	unit_id	uuid	NO	—	—	Yes	—	Unit id identifier.	Source-system Bronze field; no analytical transformation.	—
10	active	boolean	NO	true	—	No/Not shown	—	Active.	Source-system Bronze field; no analytical trans-	—

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
									formation.	
11	created_by	uuid	NO	—	—	Yes	—	Created by.	Source-system Bronze field; no analytical transformation.	—
12	created_on	timestamp with time zone	NO	now()	—	No/Not shown	—	Created on.	Source-system Bronze field; no analytical transformation.	—
13	meter_sn	character varying	YES	—	—	No/Not shown	—	Meter sn.	Source-system Bronze field; no analytical transformation.	—
14	meter_label	character varying	YES	—	—	No/Not shown	—	Canonical human-readable meter label.	Source-system Bronze field; no analytical transformation.	—

19.2 `bronze.telemetries`

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
1	device_id	uuid	NO	—	PK	Yes	—	Device id identifier.	Source-system Bronze field; no analytical transformation.	—
2	key_id	bigint	NO	nextval('bronze.key_id_seq'::regclass)	PK	Yes	—	Key id identifier.	Source-system Bronze field; no analytical trans-	—

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
									formation.	
3	timestamp	bigint	NO	—	PK	Yes	—	Timestamp.	Source-system Bronze field; no analytical transformation.	—
4	bool	boolean	YES	—	—	No/Not shown	—	Bool.	Source-system Bronze field; no analytical transformation.	—
5	dbl	double precision	YES	—	—	No/Not shown	—	Dbl.	Source-system Bronze field; no analytical transformation.	—
6	json	json	YES	—	—	No/Not shown	—	Json.	Source-system Bronze field; no analytical transformation.	—
7	long	bigint	YES	—	—	No/Not shown	—	Long.	Source-system Bronze field; no analytical transformation.	—
8	str	character varying	YES	—	—	No/Not shown	—	Str.	Source-system Bronze field; no analytical transformation.	—

19.3 `bronze.telemetries_dictionaries`

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
1	id	bigint	NO	nextval('bronze.id_seq'::regclass)	PK	Yes	—	Id.	Source-system Bronze field; no analytical transformation.	—
2	key	character varying	NO	—	—	Yes	—	Key.	Source-system Bronze field; no analytical transformation.	—
3	created_on	timestamp without time zone	NO	now()	—	No/Not shown	—	Created on.	Source-system Bronze field; no analytical transformation.	—

19.4 `gold.billing_account`

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
1	grid_billing_account_key	uuid	YES	—	—	No/Not shown	—	Grid billing-account identifier associated with the record.	Pass-through from `silver.dim_grid_billing_account_staging.grid_billing_account_key`.	—
2	facility_key	uuid	YES	—	—	No/Not shown	—	Facility identifier associated with the record.	Pass-through from `silver.dim_grid_b`	—

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
									`illing_account_staging.facility_key`.	
3	zone_key	uuid	YES	—	—	No/Not shown	—	Zone/unit identifier associated with the record.	Pass-through from `silver.dim_grid_billing_account_staging.zone_key`.	—
4	account_label	text	YES	—	—	No/Not shown	—	Account label.	Pass-through from `silver.dim_grid_billing_account_staging.account_label`.	—
5	account_number	text	YES	—	—	No/Not shown	—	Account number.	Pass-through from `silver.dim_grid_billing_account_staging.account_number`.	—
6	meter_serial_number	text	YES	—	—	No/Not shown	—	Meter serial number.	Pass-through from `silver.dim_grid_billing_account_staging.me`	—

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
									`ter_serial_number`.	
7	account_type_key	uuid	YES	—	—	No/Not shown	—	Account type key identifier.	Pass-through from `silver.dim_grid_billing_account_staging.account_type_key`.	—
8	account_type_name	text	YES	—	—	No/Not shown	—	Account type name.	Pass-through from `silver.dim_grid_billing_account_staging.account_type_name`.	—
9	service_class_key	uuid	YES	—	—	No/Not shown	—	Service class key identifier.	Pass-through from `silver.dim_grid_billing_account_staging.service_class_key`.	—
10	service_class_name	text	YES	—	—	No/Not shown	—	Service class name.	Pass-through from `silver.dim_grid_billing_account_staging.service_c`	—

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
									`lass_name`.	
11	breaker_capacity	integer	YES	—	—	No/Not shown	—	Breaker capacity.	Pass-through from `silver.dim_grid_billing_account_staging.breaker_capacity`.	—
12	minimal_breaker_capacity	integer	YES	—	—	No/Not shown	—	Minimal breaker capacity.	Pass-through from `silver.dim_grid_billing_account_staging.minimal_breaker_capacity`.	—
13	maximum_breaker_capacity	integer	YES	—	—	No/Not shown	—	Maximum breaker capacity.	Pass-through from `silver.dim_grid_billing_account_staging.maximum_breaker_capacity`.	—
14	service_class_fee	integer	YES	—	—	No/Not shown	—	Service class fee.	Pass-through from `silver.dim_grid_billing_account_sta-	—

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
									`ging.service_class_fee`.	
15	multiplication_factor	integer	YES	—	—	No/Not shown	—	Multiplication factor.	Pass-through from `silver.dim_grid_billing_account_staging.multiplication_factor`.	—
16	start_period	timestamp with time zone	YES	—	—	No/Not shown	—	Start period.	Pass-through from `silver.dim_grid_billing_account_staging.start_period`.	—
17	end_period	timestamp with time zone	YES	—	—	No/Not shown	—	End period.	Pass-through from `silver.dim_grid_billing_account_staging.end_period`.	—
18	configured_billing_days	integer	YES	—	—	No/Not shown	—	Configured billing days.	Pass-through from `silver.dim_grid_billing_account_staging.co`	—

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
									`nfigured_billing_days`.	
19	source_created_at	timestamp with time zone	YES	—	—	No/Not shown	—	Source created at.	Pass-through from `silver.dim_grid_billing_account_staging.source_created_at`.	—

19.5 `gold.date`

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
1	date_key	integer	YES	—	—	No/Not shown	—	Calendar date surrogate/key in YYYYMMDD-style integer form.	Pass-through from `silver.dim_date_staging.date_key`.	—
2	calendar_date	date	YES	—	—	No/Not shown	—	Calendar date.	Pass-through from `silver.dim_date_staging.calendar_date`.	—
3	calendar_year	integer	YES	—	—	No/Not shown	—	Calendar year.	Pass-through from `silver.dim_date_staging.calen-`	—

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
									`dar_year`.	
4	calendar_quarter	integer	YES	—	—	No/Not shown	—	Calendar quarter.	Pass-through from `silver.dim_date_staging.calendar_quarter`.	—
5	month_number	integer	YES	—	—	No/Not shown	—	Month number.	Pass-through from `silver.dim_date_staging.month_number`.	—
6	month_name	text	YES	—	—	No/Not shown	—	Month name.	Pass-through from `silver.dim_date_staging.month_name`.	—
7	month_short_name	text	YES	—	—	No/Not shown	—	Month short name.	Pass-through from `silver.dim_date_staging.month_short_name`.	—
8	week_of_year	integer	YES	—	—	No/Not shown	—	Week of year.	Pass-through from `silver.dim_date_staging.week_of_year`.	—
9		integer	YES	—	—		—		Pass-through	—

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
	iso_day_of_week					No/Not shown		Iso day of week.	from `silver.dim_date_staging.iso_day_of_week`.	
10	day_name	text	YES	—	—	No/Not shown	—	Day name.	Pass-through from `silver.dim_date_staging.day_name`.	—
11	day_short_name	text	YES	—	—	No/Not shown	—	Day short name.	Pass-through from `silver.dim_date_staging.day_short_name`.	—
12	day_of_month	integer	YES	—	—	No/Not shown	—	Day of month.	Pass-through from `silver.dim_date_staging.day_of_month`.	—
13	day_of_year	integer	YES	—	—	No/Not shown	—	Day of year.	Pass-through from `silver.dim_date_staging.day_of_year`.	—
14	is_weekend	boolean	YES	—	—	No/Not shown	—	Is weekend flag.	Pass-through from `silver.dim_date_staging.is`	—

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
									`_weekend` .	
15	month_start_date	date	YES	—	—	No/Not shown	—	Month start date.	Pass-through from `silver.dim_date_staging.month_start_date` .	—
16	month_end_date	date	YES	—	—	No/Not shown	—	Month end date.	Pass-through from `silver.dim_date_staging.month_end_date` .	—
17	quarter_start_date	date	YES	—	—	No/Not shown	—	Quarter start date.	Pass-through from `silver.dim_date_staging.quarter_start_date` .	—
18	year_start_date	date	YES	—	—	No/Not shown	—	Year start date.	Pass-through from `silver.dim_date_staging.year_start_date` .	—

19.6 `gold.energy_bill`

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
1	energy_bill_key	uuid	YES	—	—	No/Not shown	—	Energy bill key identifier.	Pass-through from `sil-`	—

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
									`ver.fact_energy_bill_staging.energy_bill_key`.	
2	facility_key	uuid	YES	—	—	No/Not shown	—	Facility identifier associated with the record.	Pass-through from `silver.fact_energy_bill_staging.facility_key`.	—
3	zone_key	uuid	YES	—	—	No/Not shown	—	Zone/unit identifier associated with the record.	Pass-through from `silver.fact_energy_bill_staging.zone_key`.	—
4	grid_billing_account_key	uuid	YES	—	—	No/Not shown	—	Grid billing-account identifier associated with the record.	Pass-through from `silver.fact_energy_bill_staging.grid_billing_account_key`.	—
5	start_date_key	integer	YES	—	—	No/Not shown	—	Start date key identifier.	Pass-through from `silver.fact_energy_bill_staging.start_date_key`.	—
6	end_date_key	integer	YES	—	—	No/Not shown	—	End date key identifier.	Pass-through from	—

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
									`silver.fact_energy_bill_staging.end_date_key`.	
7	start_period	timestamp with time zone	YES	—	—	No/Not shown	—	Start period.	Pass-through from `silver.fact_energy_bill_staging.start_period`.	—
8	end_period	timestamp with time zone	YES	—	—	No/Not shown	—	End period.	Pass-through from `silver.fact_energy_bill_staging.end_period`.	—
9	calculated_billing_days	integer	YES	—	—	No/Not shown	—	Calculated billing days.	Pass-through from `silver.fact_energy_bill_staging.calculated_billing_days`.	—
10	invoice_number	text	YES	—	—	No/Not shown	—	Invoice number.	Pass-through from `silver.fact_energy_bill_staging.invoice_number`.	—
11		numeric	YES	—	—		—			—

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
	beginning_reading					No/Not shown		Beginning reading.	Pass-through from `silver.fact_energy_bill_staging.beginning_reading`.	
12	ending_reading	numeric	YES	—	—	No/Not shown	—	Ending reading.	Pass-through from `silver.fact_energy_bill_staging.ending_reading`.	—
13	billed_consumption_kwh	numeric	YES	—	—	No/Not shown	kWh	Billed consumption kwh.	Pass-through from `silver.fact_energy_bill_staging.billed_consumption_kwh`.	—
14	billed_cost	numeric	YES	—	—	No/Not shown	—	Billed cost.	Pass-through from `silver.fact_energy_bill_staging.billed_cost`.	—
15	system_beginning_reading	numeric	YES	—	—	No/Not shown	—	System beginning reading.	Pass-through from `silver.fact_energy_bill_staging.sy`	—

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
									`stem_beginning_reading`.	
16	system_ending_reading	numeric	YES	—	—	No/Not shown	—	System ending reading.	Pass-through from `silver.fact_energy_bill_staging.system_ending_reading`.	—
17	system_consumption_kwh	numeric	YES	—	—	No/Not shown	kWh	System consumption kwh.	Pass-through from `silver.fact_energy_bill_staging.system_consumption_kwh`.	—
18	system_estimated_cost	numeric	YES	—	—	No/Not shown	—	System estimated cost.	Pass-through from `silver.fact_energy_bill_staging.system_estimated_cost`.	—
19	consumption_variance_kwh	numeric	YES	—	—	No/Not shown	kWh	Consumption variance kwh.	Pass-through from `silver.fact_energy_bill_staging.consumption_vari-`	—

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
									`ance_kwh`.	
20	cost_variance	numeric	YES	—	—	No/Not shown	—	Cost variance.	Pass-through from `silver.fact_energy_bill_staging.cost_variance`.	—
21	consumption_variance_percent	numeric	YES	—	—	No/Not shown	%	Consumption variance percent.	Pass-through from `silver.fact_energy_bill_staging.consumption_variance_percent`.	—
22	comparing_with_last_bill	text	YES	—	—	No/Not shown	—	Comparing with last bill.	Pass-through from `silver.fact_energy_bill_staging.comparing_with_last_bill`.	—
23	comparing_with_grid_bill	text	YES	—	—	No/Not shown	—	Comparing with grid bill.	Pass-through from `silver.fact_energy_bill_staging.comparing_with_grid_bill`.	—
24	system_read	timestamp without	YES	—	—	No/Not shown		System reading	Pass-through	—

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
	ing_timestamp_utc	time zone					UTC timestamp	timestamp utc.	from `silver.fact_energy_bill_staging.system_reading_timestamp_utc`.	
25	source_created_at	timestamp with time zone	YES	—	—	No/Not shown	—	Source created at.	Pass-through from `silver.fact_energy_bill_staging.source_created_at`.	—

19.7 `gold.energy_consumption_validated`

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
1	energy_meter_key	uuid	YES	—	—	No/Not shown	—	Canonical reporting-device / energy-meter identifier used across Silver and Gold.	Pass-through from `silver.fact_energy_consumption_validated.energy_meter_key`.	—
2	facility_key	uuid	YES	—	—	No/Not shown	—	Facility identifier associated with the record.	Pass-through from `silver.fact_energy_consumption_valid-`	—

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
									`ated.facility_key`.	
3	zone_key	uuid	YES	—	—	No/Not shown	—	Zone/unit identifier associated with the record.	Pass-through from `silver.fact_energy_consumption_validated.zone_key`.	—
4	grid_billing_account_key	uuid	YES	—	—	No/Not shown	—	Grid billing-account identifier associated with the record.	Pass-through from `silver.fact_energy_consumption_validated.grid_billing_account_key`.	—
5	meter_label	text	YES	—	—	No/Not shown	—	Canonical human-readable meter label.	Pass-through from `silver.fact_energy_consumption_validated.meter_label`.	—
6	reading_timestamp_utc	timestamp without time zone	YES	—	—	No/Not shown	UTC timestamp	Telemetry reading timestamp in UTC at native source precision.	Pass-through from `silver.fact_energy_consumption_validated.reading_tim`	—

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
									`estamp_utc`.	
7	reading_timestamp_saudi	timestamp without time zone	YES	—	—	No/Not shown	Asia/Riyadh timestamp	Telemetry reading timestamp converted to Asia/Riyadh local time.	Pass-through from `silver.fact_energy_consumption_validated.reading_timestamp_saudi`.	—
8	reading_date_saudi	date	YES	—	—	No/Not shown	—	Saudi-local calendar date derived from the telemetry timestamp.	Pass-through from `silver.fact_energy_consumption_validated.reading_date_saudi`.	—
9	active_energy_cumulative_kwh	double precision	YES	—	—	No/Not shown	kWh	Cumulative active-energy register reported by the meter.	Pass-through from `silver.fact_energy_consumption_validated.active_energy_cumulative_kwh`.	DQ-KWH-001
10	previous_valid_timestamp	timestamp without time zone	YES	—	—	No/Not shown	—	Timestamp of the preceding reading retained as valid for	Pass-through from `silver.fact_energy_co`	DQ-CONS-001

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
								consumption differencing.	`nsumption_validated.previous_valid_timestamp`.	
11	previous_valid_kwh	double precision	YES	—	—	No/Not shown	kWh	Cumulative kWh value of the preceding retained valid reading.	Pass-through from `silver.fact_energy_consumption_validated.previous_valid_kwh`.	DQ-CONS-001
12	validated_kwh_consumption	double precision	YES	—	—	No/Not shown	—	Validated interval energy consumption derived from cumulative kWh.	Pass-through from `silver.fact_energy_consumption_validated.validated_kwh_consumption`.	DQ-CONS-001
13	consumption_cleaning_status	text	YES	—	—	No/Not shown	—	Consumption-cleaning decision/status assigned to the reading.	Pass-through from `silver.fact_energy_consumption_validated.consumption_cleaning_status`.	DQ-CONS-001
14		text	YES	—	—		—	Quality class as-	Pass-through	

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
	interval_quality					No/Not shown		signed to the validated consumption interval.	from `silver.fact_energy_consumption_validated.interval_quality`.	DQ-CONS-001
15	gap_minutes	numeric	YES	—	—	No/Not shown	minutes	Elapsed minutes between relevant consecutive valid/readings.	Pass-through from `silver.fact_energy_consumption_validated.gap_minutes`.	DQ-GAP-001

19.8 `gold.energy_raw`

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
1	energy_meter_key	uuid	YES	—	—	No/Not shown	—	Canonical reporting-device / energy-meter identifier used across Silver and Gold.	Pass-through from `silver.fact_energy_readings_snapshot.energy_meter_key`.	—
2	facility_key	uuid	YES	—	—	No/Not shown	—	Facility identifier associated with the record.	Pass-through from `silver.fact_energy_readings_snap-`	—

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
									`shot.facility_key`.	
3	zone_key	uuid	YES	—	—	No/Not shown	—	Zone/unit identifier associated with the record.	Pass-through from `silver.fact_energy_readings_snapshot.zone_key`.	—
4	grid_billing_account_key	uuid	YES	—	—	No/Not shown	—	Grid billing-account identifier associated with the record.	Pass-through from `silver.fact_energy_readings_snapshot.grid_billing_account_key`.	—
5	reading_timestamp_utc	timestamp without time zone	YES	—	—	No/Not shown	UTC timestamp	Telemetry reading timestamp in UTC at native source precision.	Pass-through from `silver.fact_energy_readings_snapshot.reading_timestamp_utc`.	—
6	active_energy_cumulative_kwh	double precision	YES	—	—	No/Not shown	kWh	Cumulative active-energy register reported by the meter.	Pass-through from `silver.fact_energy_readings_snap-	DQ-KWH-001

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
									`shot.active_energy_cumulative_kwh`.	
7	reactive_energy_cumulative_kvarh	double precision	YES	—	—	No/Not shown	kVArh	Electrical measurement pivoted from Bronze telemetry key `kVArh`.	Pass-through from `silver.fact_energy_readings_snapshot.reactive_energy_cumulative_kvarh`.	—
8	apparent_energy_cumulative_kvah	double precision	YES	—	—	No/Not shown	kVAh	Electrical measurement pivoted from Bronze telemetry key `kVAh`.	Pass-through from `silver.fact_energy_readings_snapshot.apparent_energy_cumulative_kvah`.	—
9	voltage_neutral_1_v	double precision	YES	—	—	No/Not shown	V	Electrical measurement pivoted from Bronze telemetry key `Vn1`.	Pass-through from `silver.fact_energy_readings_snapshot.voltage_neutral_1_v`.	DQ-VN-001
10		double precision	YES	—	—	No/Not shown	V	Electrical measurement	Pass-through from	DQ-VN-001

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
	voltage_neutral_2_v							pivoted from Bronze telemetry key `Vn2`.	`silver.fact_energy_readings_snapshot.voltage_neutral_2_v`.	
11	voltage_neutral_3_v	double precision	YES	—	—	No/Not shown	V	Electrical measurement pivoted from Bronze telemetry key `Vn3`.	Pass-through from `silver.fact_energy_readings_snapshot.voltage_neutral_3_v`.	DQ-VN-001
12	voltage_neutral_average_v	double precision	YES	—	—	No/Not shown	V	Electrical measurement pivoted from Bronze telemetry key `Vna`.	Pass-through from `silver.fact_energy_readings_snapshot.voltage_neutral_average_v`.	DQ-VN-001
13	voltage_line_1_v	double precision	YES	—	—	No/Not shown	V	Electrical measurement pivoted from Bronze telemetry key `V1`.	Pass-through from `silver.fact_energy_readings_snapshot.voltage_line_1_v`.	DQ-VL-001
14	voltage_line_2_v	double precision	YES	—	—	No/Not shown	V	Electrical measurement	Pass-through from	DQ-VL-001

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
								pivoted from Bronze telemetry key `V2`.	`silver.fact_energy_readings_snapshot.voltage_line_2_v`.	
15	voltage_line_3_v	double precision	YES	—	—	No/Not shown	V	Electrical measurement pivoted from Bronze telemetry key `V3`.	Pass-through from `silver.fact_energy_readings_snapshot.voltage_line_3_v`.	DQ-VL-001
16	voltage_line_average_v	double precision	YES	—	—	No/Not shown	V	Electrical measurement pivoted from Bronze telemetry key `Va`.	Pass-through from `silver.fact_energy_readings_snapshot.voltage_line_average_v`.	DQ-VL-001
17	current_phase_1_a	double precision	YES	—	—	No/Not shown	A	Electrical measurement pivoted from Bronze telemetry key `Amp1`.	Pass-through from `silver.fact_energy_readings_snapshot.current_phase_1_a`.	DQ-CUR-001
18	current_phase_2_a	double precision	YES	—	—	No/Not shown	A	Electrical measurement pivoted from	Pass-through from `silver.fac`	DQ-CUR-001

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
								Bronze telemetry key `Amp2`.	`t_energy_readings_snapshot.current_phase_2_a`.	
19	current_phase_3_a	double precision	YES	—	—	No/Not shown	A	Electrical measurement pivoted from Bronze telemetry key `Amp3`.	Pass-through from `silver.fact_energy_readings_snapshot.current_phase_3_a`.	DQ-CUR-001
20	frequency_hz	double precision	YES	—	—	No/Not shown	Hz	Electrical measurement pivoted from Bronze telemetry key `Hz`.	Pass-through from `silver.fact_energy_readings_snapshot.frequency_hz`.	DQ-FREQ-001
21	power_factor_phase_1	double precision	YES	—	—	No/Not shown	unitless	Electrical measurement pivoted from Bronze telemetry key `PF1`.	Pass-through from `silver.fact_energy_readings_snapshot.power_factor_phase_1`.	DQ-PF-001
22	power_factor_phase_2	double precision	YES	—	—	No/Not shown	unitless	Electrical measurement pivoted from Bronze	Pass-through from `silver.fact_en-	DQ-PF-001

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
								telemetry key `PF2`.	`ergy_readings_snapshot.power_factor_phase_2`.	
23	power_factor_phase_3	double precision	YES	—	—	No/Not shown	unitless	Electrical measurement pivoted from Bronze telemetry key `PF3`.	Pass-through from `silver.fact_energy_readings_snapshot.power_factor_phase_3`.	DQ-PF-001
24	power_factor_average	double precision	YES	—	—	No/Not shown	unitless	Electrical measurement pivoted from Bronze telemetry key `PFa`.	Pass-through from `silver.fact_energy_readings_snapshot.power_factor_average`.	DQ-PF-001
25	active_power_phase_1_kw	double precision	YES	—	—	No/Not shown	kW	Electrical measurement pivoted from Bronze telemetry key `kW1`.	Pass-through from `silver.fact_energy_readings_snapshot.active_power_phase_1_kw`.	—
26	active_power_phase_2_kw	double precision	YES	—	—	No/Not shown	kW	Electrical measurement pivoted from Bronze	Pass-through from `silver.fact_en-	—

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
								telemetry key `kW2`.	`ergy_readings_snapshot.active_power_phase_2_kw`.	
27	active_power_phase_3_kw	double precision	YES	—	—	No/Not shown	kW	Electrical measurement pivoted from Bronze telemetry key `kW3`.	Pass-through from `silver.fact_energy_readings_snapshot.active_power_phase_3_kw`.	—
28	active_power_total_kw	double precision	YES	—	—	No/Not shown	kW	Electrical measurement pivoted from Bronze telemetry key `TkW`.	Pass-through from `silver.fact_energy_readings_snapshot.active_power_total_kw`.	—
29	apparent_power_phase_1_kva	double precision	YES	—	—	No/Not shown	kVA	Electrical measurement pivoted from Bronze telemetry key `kVA1`.	Pass-through from `silver.fact_energy_readings_snapshot.apparent_power_phase_1_kva`.	—
30	apparent_power_phase_2_kva	double precision	YES	—	—	No/Not shown	kVA	Electrical measurement pivoted from	Pass-through from `silver.fac`	—

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
								Bronze telemetry key `kVA2`.	`t_energy_readings_snapshot.apparent_power_phase_2_kva`.	
31	apparent_power_phase_3_kva	double precision	YES	—	—	No/Not shown	kVA	Electrical measurement pivoted from Bronze telemetry key `kVA3`.	Pass-through from `silver.fact_energy_readings_snapshot.apparent_power_phase_3_kva`.	—
32	apparent_power_total_kva	double precision	YES	—	—	No/Not shown	kVA	Electrical measurement pivoted from Bronze telemetry key `TkVA`.	Pass-through from `silver.fact_energy_readings_snapshot.apparent_power_total_kva`.	—
33	reactive_power_phase_1_kvar	double precision	YES	—	—	No/Not shown	kVAr	Electrical measurement pivoted from Bronze telemetry key `kVAr1`.	Pass-through from `silver.fact_energy_readings_snapshot.reactive_power_phase_1_kvar`.	—

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
34	reactive_power_phase_2_kvar	double precision	YES	—	—	No/Not shown	kVAr	Electrical measurement pivoted from Bronze telemetry key `kVAr2`.	Pass-through from `silver.fact_energy_readings_snapshot.reactive_power_phase_2_kvar`.	—
35	reactive_power_phase_3_kvar	double precision	YES	—	—	No/Not shown	kVAr	Electrical measurement pivoted from Bronze telemetry key `kVAr3`.	Pass-through from `silver.fact_energy_readings_snapshot.reactive_power_phase_3_kvar`.	—
36	reactive_power_total_kvar	double precision	YES	—	—	No/Not shown	kVAr	Electrical measurement pivoted from Bronze telemetry key `TkVAr`.	Pass-through from `silver.fact_energy_readings_snapshot.reactive_power_total_kvar`.	—
37	status_vn1	text	YES	—	—	No/Not shown	—	DQ status for Phase-1 line-to-neutral voltage.	Pass-through from `silver.fact_energy_readings_sn`	DQ-VN-001

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
									`apshot.status_vn1`.	
38	status_vn2	text	YES	—	—	No/Not shown	—	DQ status for Phase-2 line-to-neutral voltage.	Pass-through from `silver.fact_energy_readings_snapshot.status_vn2`.	DQ-VN-001
39	status_vn3	text	YES	—	—	No/Not shown	—	DQ status for Phase-3 line-to-neutral voltage.	Pass-through from `silver.fact_energy_readings_snapshot.status_vn3`.	DQ-VN-001
40	status_v1	text	YES	—	—	No/Not shown	—	DQ status for Phase-1 line voltage.	Pass-through from `silver.fact_energy_readings_snapshot.status_v1`.	DQ-VL-001
41	status_v2	text	YES	—	—	No/Not shown	—	DQ status for Phase-2 line voltage.	Pass-through from `silver.fact_energy_readings_snapshot.status_v2`.	DQ-VL-001
42	status_v3	text	YES	—	—		—			

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
						No/Not shown		DQ status for Phase-3 line voltage.	Pass-through from `silver.fact_energy_readings_snapshot.status_v3`.	DQ-VL-001
43	status_amp1	text	YES	—	—	No/Not shown	—	Production current-classifier result for Phase 1.	Pass-through from `silver.fact_energy_readings_snapshot.status_amp1`.	DQ-CUR-001
44	status_amp2	text	YES	—	—	No/Not shown	—	Production current-classifier result for Phase 2.	Pass-through from `silver.fact_energy_readings_snapshot.status_amp2`.	DQ-CUR-001
45	status_amp3	text	YES	—	—	No/Not shown	—	Production current-classifier result for Phase 3.	Pass-through from `silver.fact_energy_readings_snapshot.status_amp3`.	DQ-CUR-001
46	status_hz	text	YES	—	—	No/Not shown	Hz	DQ status for frequency.	Pass-through from `silver.fac`	DQ-FREQ-001

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
									`t_energy_readings_snapshot.status_hz`.	
47	status_pf1	text	YES	—	—	No/Not shown	—	DQ status for Phase-1 power factor.	Pass-through from `silver.fact_energy_readings_snapshot.status_pf1`.	DQ-PF-001
48	status_pf2	text	YES	—	—	No/Not shown	—	DQ status for Phase-2 power factor.	Pass-through from `silver.fact_energy_readings_snapshot.status_pf2`.	DQ-PF-001
49	status_pf3	text	YES	—	—	No/Not shown	—	DQ status for Phase-3 power factor.	Pass-through from `silver.fact_energy_readings_snapshot.status_pf3`.	DQ-PF-001
50	status_pfa	text	YES	—	—	No/Not shown	—	DQ status for average power factor.	Pass-through from `silver.fact_energy_readings_snap-`	DQ-PF-001

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
									`shot.status_pfa`.	
51	status_kwh	text	YES	—	—	No/Not shown	kWh	DQ status for cumulative active-energy direction.	Pass-through from `silver.fact_energy_readings_snapshot.status_kwh`.	DQ-KWH-001
52	overall_status	text	YES	—	—	No/Not shown	—	Worst status across the 12 status inputs included by the live `worst_status()` call; current statuses are excluded.	Pass-through from `silver.fact_energy_readings_snapshot.overall_status`.	—

19.9 `gold.energy_reading_gaps`

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
1	energy_meter_key	uuid	YES	—	—	No/Not shown	—	Canonical reporting-device / energy-meter identifier used across Silver and Gold.	Derived physical gap table from consecutive analytical meter-reading timestamps.	—
2	meter_label	text	YES	—	—	No/Not shown	—	Canonical human-readable	Derived physical gap table from consecutive	—

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
								meter label.	analytical meter-reading timestamps.	
3	previous_reading_timestamp	timestamp without time zone	YES	—	—	No/Not shown	—	Previous reading timestamp.	Derived physical gap table from consecutive analytical meter-reading timestamps.	DQ-GAP-001
4	current_reading_timestamp	timestamp without time zone	YES	—	—	No/Not shown	—	Current reading timestamp.	Derived physical gap table from consecutive analytical meter-reading timestamps.	DQ-GAP-001
5	gap_seconds	numeric	YES	—	—	No/Not shown	seconds	Elapsed seconds between consecutive meter readings for a recorded gap.	Derived physical gap table from consecutive analytical meter-reading timestamps.	DQ-GAP-001
6	gap_minutes	numeric	YES	—	—	No/Not shown	minutes	Elapsed minutes between relevant consecutive valid/readings.	Derived physical gap table from consecutive analytical meter-reading timestamps.	DQ-GAP-001
7	gap_hours	numeric	YES	—	—	No/Not shown	hours	Elapsed hours between consecutive meter readings for a recorded gap.	Derived physical gap table from consecutive analytical meter-reading timestamps.	DQ-GAP-001
8	gap_days	numeric	YES	—	—		days			

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
						No/Not shown		Elapsed days between consecutive meter readings for a recorded gap.	Derived physical gap table from consecutive analytical meter-reading timestamps.	DQ-GAP-001

19.10 `gold.facility`

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
1	facility_key	uuid	YES	—	—	No/Not shown	—	Facility identifier associated with the record.	Pass-through from `silver.dim_facility_staging.facility_key`.	—
2	facility_name	text	YES	—	—	No/Not shown	—	Facility name.	Pass-through from `silver.dim_facility_staging.facility_name`.	—
3	facility_type_key	uuid	YES	—	—	No/Not shown	—	Facility type key identifier.	Pass-through from `silver.dim_facility_staging.facility_type_key`.	—
4	facility_type_name	text	YES	—	—	No/Not shown	—	Facility type name.	Pass-through from `silver.dim_facil-`	—

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
									`ity_staging.facility_type_name`.	
5	latitude	numeric	YES	—	—	No/Not shown	decimal degrees	Latitude.	Pass-through from `silver.dim_facility_staging.latitude`.	—
6	longitude	numeric	YES	—	—	No/Not shown	decimal degrees	Longitude.	Pass-through from `silver.dim_facility_staging.longitude`.	—
7	x_coordinate	numeric	YES	—	—	No/Not shown	—	X coordinate.	Pass-through from `silver.dim_facility_staging.x_coordinate`.	—
8	y_coordinate	numeric	YES	—	—	No/Not shown	—	Y coordinate.	Pass-through from `silver.dim_facility_staging.y_coordinate`.	—
9	address_key	uuid	YES	—	—	No/Not shown	—	Address key identifier.	Pass-through from `silver.dim_facility_staging.address_key`.	—

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
10	address_title	text	YES	—	—	No/Not shown	—	Address title.	Pass-through from `silver.dim_facility_staging.address_title`.	—
11	short_address	text	YES	—	—	No/Not shown	—	Short address.	Pass-through from `silver.dim_facility_staging.short_address`.	—
12	building_number	integer	YES	—	—	No/Not shown	—	Building number.	Pass-through from `silver.dim_facility_staging.building_number`.	—
13	street_name	text	YES	—	—	No/Not shown	—	Street name.	Pass-through from `silver.dim_facility_staging.street_name`.	—
14	district_name	text	YES	—	—	No/Not shown	—	District name.	Pass-through from `silver.dim_facility_staging.district_name`.	—
15	city_name	text	YES	—	—	No/Not shown	—	City name.	Pass-through from `sil-	

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
									`ver.dim_facility_staging.city_name`.	
16	zip_code	integer	YES	—	—	No/Not shown	—	Zip code.	Pass-through from `silver.dim_facility_staging.zip_code`.	—
17	additional_number	integer	YES	—	—	No/Not shown	—	Additional number.	Pass-through from `silver.dim_facility_staging.additional_number`.	—
18	unit_number	integer	YES	—	—	No/Not shown	—	Unit number.	Pass-through from `silver.dim_facility_staging.unit_number`.	—
19	source_created_at	timestamp without time zone	YES	—	—	No/Not shown	—	Source created at.	Pass-through from `silver.dim_facility_staging.source_created_at`.	—

19.11 `gold.mapping_audit`

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
1		uuid	YES	—	—		—			—

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
	source_energy_meter_id					No/Not shown		Source `energy_meters.id` identifier retained for traceability.	Pass-through from `silver.energy_meter_mapping_audit_staging.source_energy_meter_id`.	
2	device_id	uuid	YES	—	—	No/Not shown	—	Device id identifier.	Pass-through from `silver.energy_meter_mapping_audit_staging.device_id`.	—
3	facility_id	uuid	YES	—	—	No/Not shown	—	Facility id identifier.	Pass-through from `silver.energy_meter_mapping_audit_staging.facility_id`.	—
4	zone_key	uuid	YES	—	—	No/Not shown	—	Zone/unit identifier associated with the record.	Pass-through from `silver.energy_meter_mapping_audit_staging.zone_key`.	—
5	zone_name	text	YES	—	—	No/Not shown	—	Zone name.	Pass-through from `silver.energy_meter_mapping_audit_sta`	—

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
									`ging.zone_name`.	
6	source_device_name	text	YES	—	—	No/Not shown	—	Source device name.	Pass-through from `silver.energy_meter_mapping_audit_staging.source_device_name`.	—
7	source_meter_label	text	YES	—	—	No/Not shown	—	Source meter label.	Pass-through from `silver.energy_meter_mapping_audit_staging.source_meter_label`.	—
8	canonical_device_name	text	YES	—	—	No/Not shown	—	Canonical device name.	Pass-through from `silver.energy_meter_mapping_audit_staging.canonical_device_name`.	—
9	canonical_meter_label	text	YES	—	—	No/Not shown	—	Canonical meter label.	Pass-through from `silver.energy_meter_mapping_audit_staging.canonical_meter_label`.	—

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
10	source_meter_records_for_device	bigint	YES	—	—	No/Not shown	—	Source meter records for device.	Pass-through from `silver.energy_meter_mapping_audit_staging.source_meter_records_for_device`.	DQ-MAP-001
11	source_records_for_meter_label	bigint	YES	—	—	No/Not shown	—	Source records for meter label.	Pass-through from `silver.energy_meter_mapping_audit_staging.source_records_for_meter_label`.	DQ-MAP-001
12	business_rule_code	text	YES	—	—	No/Not shown	—	Business rule code.	Pass-through from `silver.energy_meter_mapping_audit_staging.business_rule_code`.	DQ-MAP-001
13	canonical_mapping_rank	bigint	YES	—	—	No/Not shown	—	Canonical mapping rank.	Pass-through from `silver.energy_meter_mapping_audit_staging.canonical_map-	DQ-MAP-001

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
									`ping_rank`.	
14	mapping_decision	text	YES	—	—	No/Not shown	—	Mapping decision.	Pass-through from `silver.energy_meter_mapping_audit_staging.mapping_decision`.	DQ-MAP-001
15	mapping_status	text	YES	—	—	No/Not shown	—	Mapping status controlled status field.	Pass-through from `silver.energy_meter_mapping_audit_staging.mapping_status`.	DQ-MAP-001
16	is_meter_active	boolean	YES	—	—	No/Not shown	—	Is meter active flag.	Pass-through from `silver.energy_meter_mapping_audit_staging.is_meter_active`.	—
17	meter_source_created_at	timestamp with time zone	YES	—	—	No/Not shown	—	Meter source created at.	Pass-through from `silver.energy_meter_mapping_audit_staging.meter_source_created_at`.	—

19.12 `gold.meter`

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
1	energy_meter_key	uuid	YES	—	—	No/Not shown	—	Canonical reporting-device / energy-meter identifier used across Silver and Gold.	Pass-through from `silver.dim_energy_meter_staging.energy_meter_key`.	—
2	source_energy_meter_id	uuid	YES	—	—	No/Not shown	—	Source `energy_meters.id` identifier retained for traceability.	Pass-through from `silver.dim_energy_meter_staging.source_energy_meter_id`.	—
3	facility_key	uuid	YES	—	—	No/Not shown	—	Facility identifier associated with the record.	Pass-through from `silver.dim_energy_meter_staging.facility_key`.	—
4	zone_key	uuid	YES	—	—	No/Not shown	—	Zone/unit identifier associated with the record.	Pass-through from `silver.dim_energy_meter_staging.zone_key`.	—
5	grid_billing_account_key	uuid	YES	—	—	No/Not shown	—	Grid billing-account identifier associated with the record.	Pass-through from `silver.dim_energy_meter_sta`	—

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
									`ging.grid_billing_account_key`.	
6	parent_energy_meter_source_id	uuid	YES	—	—	No/Not shown	—	Parent energy meter source id identifier.	Pass-through from `silver.dim_energy_meter_staging.parent_energy_meter_source_id`.	—
7	meter_label	text	YES	—	—	No/Not shown	—	Canonical human-readable meter label.	Pass-through from `silver.dim_energy_meter_staging.meter_label`.	—
8	meter_serial_number	text	YES	—	—	No/Not shown	—	Meter serial number.	Pass-through from `silver.dim_energy_meter_staging.meter_serial_number`.	—
9	phase_count	integer	YES	—	—	No/Not shown	—	Phase count.	Pass-through from `silver.dim_energy_meter_staging.phase_count`.	—
10	maximum_kwh	integer	YES	—	—	No/Not shown	kWh	Maximum kwh.	Pass-through from	—

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
									`silver.dim_energy_meter_staging.maximum_kwh`.	
11	energy_meter_type_key	uuid	YES	—	—	No/Not shown	—	Energy meter type key identifier.	Pass-through from `silver.dim_energy_meter_staging.energy_meter_type_key`.	—
12	energy_meter_type_name	text	YES	—	—	No/Not shown	—	Energy meter type name.	Pass-through from `silver.dim_energy_meter_staging.energy_meter_type_name`.	—
13	is_meter_active	boolean	YES	—	—	No/Not shown	—	Is meter active flag.	Pass-through from `silver.dim_energy_meter_staging.is_meter_active`.	—
14	device_name	text	YES	—	—	No/Not shown	—	Device name.	Pass-through from `silver.dim_energy_meter_staging.device_name`.	—

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
15	device_label	text	YES	—	—	No/Not shown	—	Device label.	Pass-through from `silver.dim_energy_meter_staging.device_label`.	—
16	device_serial_number	bigint	YES	—	—	No/Not shown	—	Device serial number.	Pass-through from `silver.dim_energy_meter_staging.device_serial_number`.	—
17	device_type_key	uuid	YES	—	—	No/Not shown	—	Device type key identifier.	Pass-through from `silver.dim_energy_meter_staging.device_type_key`.	—
18	device_type_name	text	YES	—	—	No/Not shown	—	Device type name.	Pass-through from `silver.dim_energy_meter_staging.device_type_name`.	—
19	device_category_key	uuid	YES	—	—	No/Not shown	—	Device category key identifier.	Pass-through from `silver.dim_energy_meter_staging.device_c-at-`	—

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
									`egory_key`.	
20	device_category_name	text	YES	—	—	No/Not shown	—	Device category name.	Pass-through from `silver.dim_energy_meter_staging.device_category_name`.	—
21	gateway_id	uuid	YES	—	—	No/Not shown	—	Gateway id identifier.	Pass-through from `silver.dim_energy_meter_staging.gateway_id`.	—
22	is_device_active	boolean	YES	—	—	No/Not shown	—	Is device active flag.	Pass-through from `silver.dim_energy_meter_staging.is_device_active`.	—
23	is_device_online	boolean	YES	—	—	No/Not shown	—	Is device online flag.	Pass-through from `silver.dim_energy_meter_staging.is_device_online`.	—
24	assembly_date	timestamp without time zone	YES	—	—	No/Not shown	—	Assembly date.	Pass-through from `silver.dim_energy_me`	—

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
									`ter_staging.assembly_date`.	
25	device_properties	jsonb	YES	—	—	No/Not shown	—	Device properties.	Pass-through from `silver.dim_energy_meter_staging.device_properties`.	—
26	source_meter_records_for_device	bigint	YES	—	—	No/Not shown	—	Source meter records for device.	Pass-through from `silver.dim_energy_meter_staging.source_meter_records_for_device`.	DQ-MAP-001
27	source_records_for_meter_label	bigint	YES	—	—	No/Not shown	—	Source records for meter label.	Pass-through from `silver.dim_energy_meter_staging.source_records_for_meter_label`.	DQ-MAP-001
28	meter_label_matches_device_name	boolean	YES	—	—	No/Not shown	—	Meter label matches device name.	Pass-through from `silver.dim_energy_meter_staging.meter_label_matches_de`	DQ-MAP-001

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
									`vice_name`.	
29	mapping_status	text	YES	—	—	No/Not shown	—	Mapping status controlled status field.	Pass-through from `silver.dim_energy_meter_staging.mapping_status`.	DQ-MAP-001
30	meter_source_created_at	timestamp with time zone	YES	—	—	No/Not shown	—	Meter source created at.	Pass-through from `silver.dim_energy_meter_staging.meter_source_created_at`.	—

19.13 `gold.quality_summary`

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
1	energy_meter_key	uuid	YES	—	—	No/Not shown	—	Canonical reporting-device / energy-meter identifier used across Silver and Gold.	Pass-through from `silver.data_quality_summary.energy_meter_key`.	—
2	meter_label	text	YES	—	—	No/Not shown	—	Canonical human-readable meter label.	Pass-through from `silver.data_quality_summary.meter_label`.	—

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
3	total_readings	integer	YES	—	—	No/Not shown	—	Total readings.	Pass-through from `silver.data_quality_summary.total_readings`.	—
4	overall_acceptable_count	integer	YES	—	—	No/Not shown	—	Overall acceptable count.	Pass-through from `silver.data_quality_summary.overall_acceptable_count`.	—
5	overall_review_required_count	integer	YES	—	—	No/Not shown	—	Overall review required count.	Pass-through from `silver.data_quality_summary.overall_review_required_count`.	—
6	overall_anomaly_count	integer	YES	—	—	No/Not shown	—	Overall anomaly count.	Pass-through from `silver.data_quality_summary.overall_anomaly_count`.	—
7	overall_problematic_count	integer	YES	—	—	No/Not shown	—	Overall problematic count.	Pass-through from `silver.data_quality_summary.overall_problem-`	—

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
									`atic_count`.	
8	overall_critical_count	integer	YES	—	—	No/Not shown	—	Overall critical count.	Pass-through from `silver.data_quality_summary.overall_critical_count`.	—
9	overall_invalid_count	integer	YES	—	—	No/Not shown	—	Overall invalid count.	Pass-through from `silver.data_quality_summary.overall_invalid_count`.	—
10	computed_at	timestamp without time zone	YES	—	—	No/Not shown	—	Computed at.	Pass-through from `silver.data_quality_summary.computed_at`.	—

19.14 `gold.zone`

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
1	zone_key	uuid	YES	—	—	No/Not shown	—	Zone/unit identifier associated with the record.	Pass-through from `silver.dim_zone_staging.zone_key`.	—
2	facility_key	uuid	YES	—	—	No/Not shown	—	Facility identifier	Pass-through	—

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
								associated with the record.	from `silver.dim_zone_staging.facility_key`.	
3	parent_zone_key	uuid	YES	—	—	No/Not shown	—	Parent zone key identifier.	Pass-through from `silver.dim_zone_staging.parent_zone_key`.	—
4	parent_zone_name	text	YES	—	—	No/Not shown	—	Parent zone name.	Pass-through from `silver.dim_zone_staging.parent_zone_name`.	—
5	zone_name	text	YES	—	—	No/Not shown	—	Zone name.	Pass-through from `silver.dim_zone_staging.zone_name`.	—
6	zone_type_key	uuid	YES	—	—	No/Not shown	—	Zone type key identifier.	Pass-through from `silver.dim_zone_staging.zone_type_key`.	—
7	zone_type_name	text	YES	—	—	No/Not shown	—	Zone type name.	Pass-through from `silver.dim_zone_staging.zo`	

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
									`ne_type_name`.	
8	hierarchy_level	integer	YES	—	—	No/Not shown	—	Hierarchy level.	Pass-through from `silver.dim_zone_staging.hierarchy_level`.	—
9	is_building	boolean	YES	—	—	No/Not shown	—	Is building flag.	Pass-through from `silver.dim_zone_staging.is_building`.	—
10	is_active	boolean	YES	—	—	No/Not shown	—	Is active flag.	Pass-through from `silver.dim_zone_staging.is_active`.	—
11	has_children	boolean	YES	—	—	No/Not shown	—	Has children flag.	Pass-through from `silver.dim_zone_staging.has_children`.	—
12	latitude	numeric	YES	—	—	No/Not shown	decimal degrees	Latitude.	Pass-through from `silver.dim_zone_staging.latitude`.	—
13	longitude	numeric	YES	—	—	No/Not shown	decimal degrees	Longitude.	Pass-through from `silver.dim`	—

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
									`_zone_staging.longitude`.	
14	x_coordinate	numeric	YES	—	—	No/Not shown	—	X coordinate.	Pass-through from `silver.dim_zone_staging.x_coordinate`.	—
15	y_coordinate	numeric	YES	—	—	No/Not shown	—	Y coordinate.	Pass-through from `silver.dim_zone_staging.y_coordinate`.	—
16	html_hex_color	text	YES	—	—	No/Not shown	—	Html hex color.	Pass-through from `silver.dim_zone_staging.html_hex_color`.	—
17	description	text	YES	—	—	No/Not shown	—	Description.	Pass-through from `silver.dim_zone_staging.description`.	—
18	source_created_at	timestamp without time zone	YES	—	—	No/Not shown	—	Source created at.	Pass-through from `silver.dim_zone_staging.source_cr`	—

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
									`eated_at`.	

19.15 `silver.dim_energy_meter_staging`

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
1	energy_meter_key	uuid	YES	—	—	No/Not shown	—	Canonical reporting-device / energy-meter identifier used across Silver and Gold.	Derived from Silver meter/device/unit/type/category landing views with canonical mapping rules.	—
2	source_energy_meter_id	uuid	YES	—	—	No/Not shown	—	Source `energy_meters.id` identifier retained for traceability.	Derived from Silver meter/device/unit/type/category landing views with canonical mapping rules.	—
3	facility_key	uuid	YES	—	—	No/Not shown	—	Facility identifier associated with the record.	Derived from Silver meter/device/unit/type/category landing views with canonical mapping rules.	—
4	zone_key	uuid	YES	—	—	No/Not shown	—	Zone/unit identifier associated with the record.	Derived from Silver meter/device/unit/type/category landing	—

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
									views with canonical mapping rules.	
5	grid_billing_account_key	uuid	YES	—	—	No/Not shown	—	Grid billing-account identifier associated with the record.	Derived from Silver meter/device/unit/type/category landing views with canonical mapping rules.	—
6	parent_energy_meter_source_id	uuid	YES	—	—	No/Not shown	—	Parent energy meter source id identifier.	Derived from Silver meter/device/unit/type/category landing views with canonical mapping rules.	—
7	meter_label	text	YES	—	—	No/Not shown	—	Canonical human-readable meter label.	Derived from Silver meter/device/unit/type/category landing views with canonical mapping rules.	—
8	meter_serial_number	text	YES	—	—	No/Not shown	—	Meter serial number.	Derived from Silver meter/device/unit/type/category landing views with canonical mapping rules.	—

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
9	phase_count	integer	YES	—	—	No/Not shown	—	Phase count.	Derived from Silver meter/device/unit/type/category landing views with canonical mapping rules.	—
10	maximum_kwh	integer	YES	—	—	No/Not shown	kWh	Maximum kwh.	Derived from Silver meter/device/unit/type/category landing views with canonical mapping rules.	—
11	energy_meter_type_key	uuid	YES	—	—	No/Not shown	—	Energy meter type key identifier.	Derived from Silver meter/device/unit/type/category landing views with canonical mapping rules.	—
12	energy_meter_type_name	text	YES	—	—	No/Not shown	—	Energy meter type name.	Derived from Silver meter/device/unit/type/category landing views with canonical mapping rules.	—
13	is_meter_active	boolean	YES	—	—	No/Not shown	—	Is meter active flag.	Derived from Silver meter/device/	—

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
									unit/type/category landing views with canonical mapping rules.	
14	device_name	text	YES	—	—	No/Not shown	—	Device name.	Derived from Silver meter/device/unit/type/category landing views with canonical mapping rules.	—
15	device_label	text	YES	—	—	No/Not shown	—	Device label.	Derived from Silver meter/device/unit/type/category landing views with canonical mapping rules.	—
16	device_serial_number	bigint	YES	—	—	No/Not shown	—	Device serial number.	Derived from Silver meter/device/unit/type/category landing views with canonical mapping rules.	—
17	device_type_key	uuid	YES	—	—	No/Not shown	—	Device type key identifier.	Derived from Silver meter/device/unit/type/category landing views with canonical	—

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
									mapping rules.	
18	device_type_name	text	YES	—	—	No/Not shown	—	Device type name.	Derived from Silver meter/device/unit/type/category landing views with canonical mapping rules.	—
19	device_category_key	uuid	YES	—	—	No/Not shown	—	Device category key identifier.	Derived from Silver meter/device/unit/type/category landing views with canonical mapping rules.	—
20	device_category_name	text	YES	—	—	No/Not shown	—	Device category name.	Derived from Silver meter/device/unit/type/category landing views with canonical mapping rules.	—
21	gateway_id	uuid	YES	—	—	No/Not shown	—	Gateway id identifier.	Derived from Silver meter/device/unit/type/category landing views with canonical mapping rules.	—
22	is_device_active	boolean	YES	—	—	No/Not shown	—	Is device active flag.	Derived from Silver	—

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
									meter/ device/ unit/type/ category landing views with canonical mapping rules.	
23	is_device_online	boolean	YES	—	—	No/Not shown	—	Is device online flag.	Derived from Silver meter/ device/ unit/type/ category landing views with canonical mapping rules.	—
24	assembly_date	timestamp without time zone	YES	—	—	No/Not shown	—	Assembly date.	Derived from Silver meter/ device/ unit/type/ category landing views with canonical mapping rules.	—
25	device_properties	jsonb	YES	—	—	No/Not shown	—	Device properties.	Derived from Silver meter/ device/ unit/type/ category landing views with canonical mapping rules.	—
26	source_meter_records_for_device	bigint	YES	—	—	No/Not shown	—	Source meter records for device.	Derived from Silver meter/ device/ unit/type/ category landing views	DQ-MAP-001

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
									with canonical mapping rules.	
27	source_records_for_meter_label	bigint	YES	—	—	No/Not shown	—	Source records for meter label.	Derived from Silver meter/device/unit/type/category landing views with canonical mapping rules.	DQ-MAP-001
28	meter_label_matches_device_name	boolean	YES	—	—	No/Not shown	—	Meter label matches device name.	Derived from Silver meter/device/unit/type/category landing views with canonical mapping rules.	DQ-MAP-001
29	mapping_status	text	YES	—	—	No/Not shown	—	Mapping status controlled status field.	Derived from Silver meter/device/unit/type/category landing views with canonical mapping rules.	DQ-MAP-001
30	meter_source_created_at	timestamp with time zone	YES	—	—	No/Not shown	—	Meter source created at.	Derived from Silver meter/device/unit/type/category landing views with canonical mapping rules.	—

19.16 `silver.dim_grid_billing_account_staging`

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
1	grid_billing_account_key	uuid	YES	—	—	No/Not shown	—	Grid billing-account identifier associated with the record.	Derived from Silver billing-account/type/service-class landing views.	—
2	facility_key	uuid	YES	—	—	No/Not shown	—	Facility identifier associated with the record.	Derived from Silver billing-account/type/service-class landing views.	—
3	zone_key	uuid	YES	—	—	No/Not shown	—	Zone/unit identifier associated with the record.	Derived from Silver billing-account/type/service-class landing views.	—
4	account_label	text	YES	—	—	No/Not shown	—	Account label.	Derived from Silver billing-account/type/service-class landing views.	—
5	account_number	text	YES	—	—	No/Not shown	—	Account number.	Derived from Silver billing-account/type/service-class landing views.	—
6	meter_serial_number	text	YES	—	—	No/Not shown	—	Meter serial number.	Derived from Silver billing-ac-	—

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
									count/ type/service-class landing views.	
7	account_type_key	uuid	YES	—	—	No/Not shown	—	Account type key identifier.	Derived from Silver billing-account/ type/service-class landing views.	—
8	account_type_name	text	YES	—	—	No/Not shown	—	Account type name.	Derived from Silver billing-account/ type/service-class landing views.	—
9	service_class_key	uuid	YES	—	—	No/Not shown	—	Service class key identifier.	Derived from Silver billing-account/ type/service-class landing views.	—
10	service_class_name	text	YES	—	—	No/Not shown	—	Service class name.	Derived from Silver billing-account/ type/service-class landing views.	—
11	breaker_capacity	integer	YES	—	—	No/Not shown	—	Breaker capacity.	Derived from Silver billing-account/ type/service-class landing views.	—

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
12	minimal_breaker_capacity	integer	YES	—	—	No/Not shown	—	Minimal breaker capacity.	Derived from Silver billing-account/type/service-class landing views.	—
13	maximum_breaker_capacity	integer	YES	—	—	No/Not shown	—	Maximum breaker capacity.	Derived from Silver billing-account/type/service-class landing views.	—
14	service_class_fee	integer	YES	—	—	No/Not shown	—	Service class fee.	Derived from Silver billing-account/type/service-class landing views.	—
15	multiplication_factor	integer	YES	—	—	No/Not shown	—	Multiplication factor.	Derived from Silver billing-account/type/service-class landing views.	—
16	start_period	timestamp with time zone	YES	—	—	No/Not shown	—	Start period.	Derived from Silver billing-account/type/service-class landing views.	—
17	end_period	timestamp with time zone	YES	—	—	No/Not shown	—	End period.	Derived from Silver billing-account/type/service-	—

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
									class landing views.	
18	configured_billing_days	integer	YES	—	—	No/Not shown	—	Configured billing days.	Derived from Silver billing-account/type/service-class landing views.	—
19	source_created_at	timestamp with time zone	YES	—	—	No/Not shown	—	Source created at.	Derived from Silver billing-account/type/service-class landing views.	—

19.17 `silver.dim_meter_cb_rating`

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
1	energy_meter_key	uuid	NO	—	PK	Yes	—	Canonical reporting-device / energy-meter identifier used across Silver and Gold.	Project-maintained breaker-rating reference table; external authoritative engineering source not embedded in catalog metadata.	—
2	cb_max_rating_a	double precision	NO	—	—	No/Not shown	A	Per-meter circuit-breaker maximum current rating used by	Project-maintained breaker-rating reference table; external authorit-	DQ-CUR-001

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
								the current classifier.	ative engineering source not embedded in catalog metadata.	
3	updated_at	timestamp without time zone	YES	now()	—	No/Not shown	—	Updated at.	Project-maintained breaker-rating reference table; external authoritative engineering source not embedded in catalog metadata.	—

19.18 `silver.energy_meter_mapping_audit_staging`

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
1	source_energy_meter_id	uuid	YES	—	—	No/Not shown	—	Source `energy_meters.id` identifier retained for traceability.	Derived from Silver meter/device/unit landing views with mapping business rules and ranking.	—
2	device_id	uuid	YES	—	—	No/Not shown	—	Device id identifier.	Derived from Silver meter/device/unit landing views with mapping business rules and ranking.	—

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
3	facility_id	uuid	YES	—	—	No/Not shown	—	Facility id identifier.	Derived from Silver meter/ device/ unit landing views with mapping business rules and ranking.	—
4	zone_key	uuid	YES	—	—	No/Not shown	—	Zone/unit identifier associated with the record.	Derived from Silver meter/ device/ unit landing views with mapping business rules and ranking.	—
5	zone_name	text	YES	—	—	No/Not shown	—	Zone name.	Derived from Silver meter/ device/ unit landing views with mapping business rules and ranking.	—
6	source_device_name	text	YES	—	—	No/Not shown	—	Source device name.	Derived from Silver meter/ device/ unit landing views with mapping business rules and ranking.	—
7	source_meter_label	text	YES	—	—	No/Not shown	—	Source meter label.	Derived from Silver meter/ device/ unit landing views with mapping	—

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
									business rules and ranking.	
8	canonical_device_name	text	YES	—	—	No/Not shown	—	Canonical device name.	Derived from Silver meter/device/unit landing views with mapping business rules and ranking.	—
9	canonical_meter_label	text	YES	—	—	No/Not shown	—	Canonical meter label.	Derived from Silver meter/device/unit landing views with mapping business rules and ranking.	—
10	source_meter_records_for_device	bigint	YES	—	—	No/Not shown	—	Source meter records for device.	Derived from Silver meter/device/unit landing views with mapping business rules and ranking.	DQ-MAP-001
11	source_records_for_meter_label	bigint	YES	—	—	No/Not shown	—	Source records for meter label.	Derived from Silver meter/device/unit landing views with mapping business rules and ranking.	DQ-MAP-001
12	business_rule_code	text	YES	—	—	No/Not shown	—	Business rule code.	Derived from Silver meter/device/unit land-	DQ-MAP-001

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
									ing views with mapping business rules and ranking.	
13	canonical_mapping_rank	bigint	YES	—	—	No/Not shown	—	Canonical mapping rank.	Derived from Silver meter/device/unit landing views with mapping business rules and ranking.	DQ-MAP-001
14	mapping_decision	text	YES	—	—	No/Not shown	—	Mapping decision.	Derived from Silver meter/device/unit landing views with mapping business rules and ranking.	DQ-MAP-001
15	mapping_status	text	YES	—	—	No/Not shown	—	Mapping status controlled status field.	Derived from Silver meter/device/unit landing views with mapping business rules and ranking.	DQ-MAP-001
16	is_meter_active	boolean	YES	—	—	No/Not shown	—	Is meter active flag.	Derived from Silver meter/device/unit landing views with mapping business rules and ranking.	—
17	meter_source_created_at	timestamp with	YES	—	—	No/Not shown	—	Meter source	Derived from Silver	—

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
		time zone						created at.	meter/ device/ unit landing views with mapping business rules and ranking.	

19.19 `silver.fact_energy_consumption_validated`

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
1	energy_meter_key	uuid	YES	—	—	Yes	—	Canonical reporting-device / energy-meter identifier used across Silver and Gold.	From reading snapshot / canonical meter dimension.	—
2	facility_key	uuid	YES	—	—	No/Not shown	—	Facility identifier associated with the record.	From reading snapshot / canonical meter dimension.	—
3	zone_key	uuid	YES	—	—	No/Not shown	—	Zone/unit identifier associated with the record.	From reading snapshot / canonical meter dimension.	—
4	grid_billing_account_key	uuid	YES	—	—	No/Not shown	—	Grid billing-account identifier associated with the record.	From reading snapshot / canonical meter dimension.	—
5	meter_label	text	YES	—	—	Yes	—	Canonical human-readable	From reading snapshot / canonical	—

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
								meter label.	meter dimension.	
6	reading_timestamp_utc	timestamp without time zone	YES	—	—	Yes	UTC timestamp	Telemetry reading timestamp in UTC at native source precision.	From reading snapshot / canonical meter dimension.	—
7	reading_timestamp_saudi	timestamp without time zone	YES	—	—	No/Not shown	Asia/Riyadh timestamp	Telemetry reading timestamp converted to Asia/Riyadh local time.	From reading snapshot / canonical meter dimension.	—
8	reading_date_saudi	date	YES	—	—	Yes	—	Saudi-local calendar date derived from the telemetry timestamp.	From reading snapshot / canonical meter dimension.	—
9	active_energy_cumulative_kwh	double precision	YES	—	—	No/Not shown	kWh	Cumulative active-energy register reported by the meter.	From reading snapshot / canonical meter dimension.	DQ-KWH-001
10	status_kwh	text	YES	—	—	No/Not shown	kWh	DQ status for cumulative active-energy direction.	From reading snapshot / canonical meter dimension.	DQ-KWH-001
11	raw_previous_kwh	double precision	YES	—	—	No/Not shown	kWh	Raw previous kwh.	Derived from cumulative-kWh validation/ differencing logic documented in §10.	DQ-CONS-001
12			YES	—	—		kWh		Derived from cu-	

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
	raw_next_kwh	double precision				No/Not shown		Raw next kwh.	mulative-kWh validation/ differencing logic documented in §10.	DQ-CONS-001
13	consumption_cleaning_status	text	YES	—	—	No/Not shown	—	Consumption-cleaning decision/ status assigned to the reading.	Derived from cumulative-kWh validation/ differencing logic documented in §10.	DQ-CONS-001
14	previous_valid_timestamp	timestamp without time zone	YES	—	—	No/Not shown	—	Timestamp of the preceding reading retained as valid for consumption differencing.	Derived from cumulative-kWh validation/ differencing logic documented in §10.	DQ-CONS-001
15	previous_valid_kwh	double precision	YES	—	—	No/Not shown	kWh	Cumulative kWh value of the preceding retained valid reading.	Derived from cumulative-kWh validation/ differencing logic documented in §10.	DQ-CONS-001
16	validated_kwh_consumption	double precision	YES	—	—	No/Not shown	—	Validated interval energy consumption derived from cumulative kWh.	Derived from cumulative-kWh validation/ differencing logic documented in §10.	DQ-CONS-001
17	gap_minutes	numeric	YES	—	—	No/Not shown	minutes	Elapsed minutes between relevant consecutive valid/readings.	Derived from cumulative-kWh validation/ differencing logic docu-	DQ-GAP-001

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
									mented in §10.	
18	interval_quality	text	YES	—	—	No/Not shown	—	Quality class assigned to the validated consumption interval.	Derived from cumulative-kWh validation/differencing logic documented in §10.	DQ-CONS-001

19.20 `silver.fact_energy_readings_flagged`

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
1	energy_meter_key	uuid	YES	—	—	No/Not shown	—	Canonical reporting-device / energy-meter identifier used across Silver and Gold.	Identity/time field passed from `fact_energy_readings_staging`.	—
2	source_energy_meter_id	uuid	YES	—	—	No/Not shown	—	Source `energy_meters.id` identifier retained for traceability.	Identity/time field passed from `fact_energy_readings_staging`.	—
3	facility_key	uuid	YES	—	—	No/Not shown	—	Facility identifier associated with the record.	Identity/time field passed from `fact_energy_readings_staging`.	—
4	zone_key	uuid	YES	—	—	No/Not shown	—	Zone/unit identifier associated with the record.	Identity/time field passed from `fact_energy_re`	—

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
									`readings_staging`.	
5	grid_billing_account_key	uuid	YES	—	—	No/Not shown	—	Grid billing-account identifier associated with the record.	Identity/time field passed from `fact_energy_readings_staging`.	—
6	date_key	integer	YES	—	—	No/Not shown	—	Calendar date surrogate/key in YYYYMMDD-style integer form.	Identity/time field passed from `fact_energy_readings_staging`.	—
7	time_key	integer	YES	—	—	No/Not shown	—	Time key derived from the reading timestamp.	Identity/time field passed from `fact_energy_readings_staging`.	—
8	source_timestamp_ms	bigint	YES	—	—	No/Not shown	—	Original Bronze telemetry epoch timestamp in milliseconds.	Identity/time field passed from `fact_energy_readings_staging`.	—
9	reading_timestamp_utc	timestamp without time zone	YES	—	—	No/Not shown	UTC timestamp	Telemetry reading timestamp in UTC at native source precision.	Identity/time field passed from `fact_energy_readings_staging`.	—
10	reading_timestamp_saudi	timestamp without time zone	YES	—	—	No/Not shown	Asia/Riyadh timestamp	Telemetry reading timestamp converted to Asia/	Identity/time field passed from `fact_energy_read-`	—

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
								Riyadh local time.	`ings_staging`.	
11	reading_date_saudi	date	YES	—	—	No/Not shown	—	Saudi-local calendar date derived from the telemetry timestamp.	Identity/time field passed from `fact_energy_readings_staging`.	—
12	status_vn1	text	YES	—	—	No/Not shown	—	DQ status for Phase-1 line-to-neutral voltage.	Derived by live Silver classifier function(s); see DQ Rule.	DQ-VN-001
13	status_vn2	text	YES	—	—	No/Not shown	—	DQ status for Phase-2 line-to-neutral voltage.	Derived by live Silver classifier function(s); see DQ Rule.	DQ-VN-001
14	status_vn3	text	YES	—	—	No/Not shown	—	DQ status for Phase-3 line-to-neutral voltage.	Derived by live Silver classifier function(s); see DQ Rule.	DQ-VN-001
15	status_v1	text	YES	—	—	No/Not shown	—	DQ status for Phase-1 line voltage.	Derived by live Silver classifier function(s); see DQ Rule.	DQ-VL-001
16	status_v2	text	YES	—	—	No/Not shown	—	DQ status for Phase-2 line voltage.	Derived by live Silver classifier function(s); see DQ Rule.	DQ-VL-001
17	status_v3	text	YES	—	—	No/Not shown	—	DQ status for Phase-3	Derived by live Silver classifier	DQ-VL-001

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
								line voltage.	function(s); see DQ Rule.	
18	status_amp1	text	YES	—	—	No/Not shown	—	Production current-classifier result for Phase 1.	Derived by live Silver classifier function(s); see DQ Rule.	DQ-CUR-001
19	status_amp2	text	YES	—	—	No/Not shown	—	Production current-classifier result for Phase 2.	Derived by live Silver classifier function(s); see DQ Rule.	DQ-CUR-001
20	status_amp3	text	YES	—	—	No/Not shown	—	Production current-classifier result for Phase 3.	Derived by live Silver classifier function(s); see DQ Rule.	DQ-CUR-001
21	status_hz	text	YES	—	—	No/Not shown	Hz	DQ status for frequency.	Derived by live Silver classifier function(s); see DQ Rule.	DQ-FREQ-001
22	status_pf1	text	YES	—	—	No/Not shown	—	DQ status for Phase-1 power factor.	Derived by live Silver classifier function(s); see DQ Rule.	DQ-PF-001
23	status_pf2	text	YES	—	—	No/Not shown	—	DQ status for Phase-2 power factor.	Derived by live Silver classifier function(s); see DQ Rule.	DQ-PF-001
24	status_pf3	text	YES	—	—	No/Not shown	—	DQ status for Phase-3	Derived by live Silver classifier	DQ-PF-001

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
								power factor.	function(s); see DQ Rule.	
25	status_pfa	text	YES	—	—	No/Not shown	—	DQ status for average power factor.	Derived by live Silver classifier function(s); see DQ Rule.	DQ-PF-001
26	status_kwh	text	YES	—	—	No/Not shown	kWh	DQ status for cumulative active-energy direction.	Derived by live Silver classifier function(s); see DQ Rule.	DQ-KWH-001
27	overall_status	text	YES	—	—	No/Not shown	—	Worst status across the 12 status inputs included by the live `worst_status()` call; current statuses are excluded.	Derived by `silver.worst_status()` over 12 non-current status inputs.	—

19.21 `silver.fact_energy_readings_snapshot`

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
1	energy_meter_key	uuid	NO	—	PK	Yes	—	Canonical reporting-device / energy-meter identifier used across Silver and Gold.	From `silver.fact_energy_readings_staging`.	—
2		uuid	YES	—	—		—			—

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
	facility_key					No/Not shown		Facility identifier associated with the record.	From `silver.fact_energy_readings_staging`.	
3	zone_key	uuid	YES	—	—	No/Not shown	—	Zone/unit identifier associated with the record.	From `silver.fact_energy_readings_staging`.	—
4	grid_billing_account_key	uuid	YES	—	—	No/Not shown	—	Grid billing-account identifier associated with the record.	From `silver.fact_energy_readings_staging`.	—
5	reading_timestamp_utc	timestamp without time zone	NO	—	PK	Yes	UTC timestamp	Telemetry reading timestamp in UTC at native source precision.	From `silver.fact_energy_readings_staging`.	—
6	active_energy_cumulative_kwh	double precision	YES	—	—	No/Not shown	kWh	Cumulative active-energy register reported by the meter.	From `silver.fact_energy_readings_staging`.	DQ-KWH-001
7	reactive_energy_cumulative_kvarh	double precision	YES	—	—	No/Not shown	kVArh	Electrical measurement pivoted from Bronze telemetry key `kVArh`.	From `silver.fact_energy_readings_staging`.	—
8	apparent_energy_cumulative_kvah	double precision	YES	—	—	No/Not shown	kVAh	Electrical measurement pivoted from Bronze	From `silver.fact_energy_readings_staging`.	—

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
								telemetry key `kVAh`.	`ings_staging`.	
9	voltage_neutral_1_v	double precision	YES	—	—	No/Not shown	V	Electrical measurement pivoted from Bronze telemetry key `Vn1`.	From `silver.fact_energy_readings_staging`.	DQ-VN-001
10	voltage_neutral_2_v	double precision	YES	—	—	No/Not shown	V	Electrical measurement pivoted from Bronze telemetry key `Vn2`.	From `silver.fact_energy_readings_staging`.	DQ-VN-001
11	voltage_neutral_3_v	double precision	YES	—	—	No/Not shown	V	Electrical measurement pivoted from Bronze telemetry key `Vn3`.	From `silver.fact_energy_readings_staging`.	DQ-VN-001
12	voltage_neutral_average_v	double precision	YES	—	—	No/Not shown	V	Electrical measurement pivoted from Bronze telemetry key `Vna`.	From `silver.fact_energy_readings_staging`.	DQ-VN-001
13	voltage_line_1_v	double precision	YES	—	—	No/Not shown	V	Electrical measurement pivoted from Bronze telemetry key `V1`.	From `silver.fact_energy_readings_staging`.	DQ-VL-001
14	voltage_line_2_v	double precision	YES	—	—	No/Not shown	V	Electrical measurement pivoted from Bronze telemetry key `V2`.	From `silver.fact_energy_readings_staging`.	DQ-VL-001
15	voltage_line_3_v	double precision	YES	—	—	No/Not shown	V	Electrical measurement	From `silver.fac`	DQ-VL-001

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
								pivoted from Bronze telemetry key `V3`.	`t_energy_readings_staging`.	
16	voltage_line_average_v	double precision	YES	—	—	No/Not shown	V	Electrical measurement pivoted from Bronze telemetry key `Va`.	From `silver.fact_energy_readings_staging`.	DQ-VL-001
17	current_phase_1_a	double precision	YES	—	—	No/Not shown	A	Electrical measurement pivoted from Bronze telemetry key `Amp1`.	From `silver.fact_energy_readings_staging`.	DQ-CUR-001
18	current_phase_2_a	double precision	YES	—	—	No/Not shown	A	Electrical measurement pivoted from Bronze telemetry key `Amp2`.	From `silver.fact_energy_readings_staging`.	DQ-CUR-001
19	current_phase_3_a	double precision	YES	—	—	No/Not shown	A	Electrical measurement pivoted from Bronze telemetry key `Amp3`.	From `silver.fact_energy_readings_staging`.	DQ-CUR-001
20	frequency_hz	double precision	YES	—	—	No/Not shown	Hz	Electrical measurement pivoted from Bronze telemetry key `Hz`.	From `silver.fact_energy_readings_staging`.	DQ-FREQ-001
21	power_factor_phase_1	double precision	YES	—	—	No/Not shown	unitless	Electrical measurement pivoted from Bronze	From `silver.fact_energy_read-`	DQ-PF-001

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
								telemetry key `PF1`.	`ings_staging`.	
22	power_factor_phase_2	double precision	YES	—	—	No/Not shown	unitless	Electrical measurement pivoted from Bronze telemetry key `PF2`.	From `silver.fact_energy_readings_staging`.	DQ-PF-001
23	power_factor_phase_3	double precision	YES	—	—	No/Not shown	unitless	Electrical measurement pivoted from Bronze telemetry key `PF3`.	From `silver.fact_energy_readings_staging`.	DQ-PF-001
24	power_factor_average	double precision	YES	—	—	No/Not shown	unitless	Electrical measurement pivoted from Bronze telemetry key `PFa`.	From `silver.fact_energy_readings_staging`.	DQ-PF-001
25	active_power_phase_1_kw	double precision	YES	—	—	No/Not shown	kW	Electrical measurement pivoted from Bronze telemetry key `kW1`.	From `silver.fact_energy_readings_staging`.	—
26	active_power_phase_2_kw	double precision	YES	—	—	No/Not shown	kW	Electrical measurement pivoted from Bronze telemetry key `kW2`.	From `silver.fact_energy_readings_staging`.	—
27	active_power_phase_3_kw	double precision	YES	—	—	No/Not shown	kW	Electrical measurement pivoted from Bronze telemetry key `kW3`.	From `silver.fact_energy_readings_staging`.	—
28	active_power_total_kw	double precision	YES	—	—	No/Not shown	kW	Electrical measurement pivoted	From `silver.fact_en-`	—

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
								from Bronze telemetry key `TkW`.	`ergy_read-ings_staging`.	
29	apparent_power_phase_1_kva	double precision	YES	—	—	No/Not shown	kVA	Electrical measurement pivoted from Bronze telemetry key `kVA1`.	From `silver.fact_energy_read-ings_staging`.	—
30	apparent_power_phase_2_kva	double precision	YES	—	—	No/Not shown	kVA	Electrical measurement pivoted from Bronze telemetry key `kVA2`.	From `silver.fact_energy_read-ings_staging`.	—
31	apparent_power_phase_3_kva	double precision	YES	—	—	No/Not shown	kVA	Electrical measurement pivoted from Bronze telemetry key `kVA3`.	From `silver.fact_energy_read-ings_staging`.	—
32	apparent_power_total_kva	double precision	YES	—	—	No/Not shown	kVA	Electrical measurement pivoted from Bronze telemetry key `TkVA`.	From `silver.fact_energy_read-ings_staging`.	—
33	reactive_power_phase_1_kvar	double precision	YES	—	—	No/Not shown	kVAr	Electrical measurement pivoted from Bronze telemetry key `kVAr1`.	From `silver.fact_energy_read-ings_staging`.	—
34	reactive_power_phase_2_kvar	double precision	YES	—	—	No/Not shown	kVAr	Electrical measurement pivoted from Bronze telemetry	From `silver.fact_energy_read-`	—

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
								key `kVAr2`.	`ings_staging`.	
35	reactive_power_phase_3_kvar	double precision	YES	—	—	No/Not shown	kVAr	Electrical measurement pivoted from Bronze telemetry key `kVAr3`.	From `silver.fact_energy_readings_staging`.	—
36	reactive_power_total_kvar	double precision	YES	—	—	No/Not shown	kVAr	Electrical measurement pivoted from Bronze telemetry key `TkVAr`.	From `silver.fact_energy_readings_staging`.	—
37	status_vn1	text	YES	—	—	No/Not shown	—	DQ status for Phase-1 line-to-neutral voltage.	From `silver.fact_energy_readings_flagged`.	DQ-VN-001
38	status_vn2	text	YES	—	—	No/Not shown	—	DQ status for Phase-2 line-to-neutral voltage.	From `silver.fact_energy_readings_flagged`.	DQ-VN-001
39	status_vn3	text	YES	—	—	No/Not shown	—	DQ status for Phase-3 line-to-neutral voltage.	From `silver.fact_energy_readings_flagged`.	DQ-VN-001
40	status_v1	text	YES	—	—	No/Not shown	—	DQ status for Phase-1 line voltage.	From `silver.fact_energy_readings_flagged`.	DQ-VL-001
41	status_v2	text	YES	—	—	No/Not shown	—	DQ status for	From `sil-`	DQ-VL-001

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
								Phase-2 line voltage.	`ver.fact_energy_readings_flagged`.	
42	status_v3	text	YES	—	—	No/Not shown	—	DQ status for Phase-3 line voltage.	From `silver.fact_energy_readings_flagged`.	DQ-VL-001
43	status_amp1	text	YES	—	—	No/Not shown	—	Production current-classifier result for Phase 1.	From `silver.fact_energy_readings_flagged`.	DQ-CUR-001
44	status_amp2	text	YES	—	—	No/Not shown	—	Production current-classifier result for Phase 2.	From `silver.fact_energy_readings_flagged`.	DQ-CUR-001
45	status_amp3	text	YES	—	—	No/Not shown	—	Production current-classifier result for Phase 3.	From `silver.fact_energy_readings_flagged`.	DQ-CUR-001
46	status_hz	text	YES	—	—	No/Not shown	Hz	DQ status for frequency.	From `silver.fact_energy_readings_flagged`.	DQ-FREQ-001
47	status_pf1	text	YES	—	—	No/Not shown	—	DQ status for Phase-1 power factor.	From `silver.fact_energy_readings_flagged`.	DQ-PF-001

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
48	status_pf2	text	YES	—	—	No/Not shown	—	DQ status for Phase-2 power factor.	From `silver.fact_energy_readings_flagged`.	DQ-PF-001
49	status_pf3	text	YES	—	—	No/Not shown	—	DQ status for Phase-3 power factor.	From `silver.fact_energy_readings_flagged`.	DQ-PF-001
50	status_pfa	text	YES	—	—	No/Not shown	—	DQ status for average power factor.	From `silver.fact_energy_readings_flagged`.	DQ-PF-001
51	status_kwh	text	YES	—	—	No/Not shown	kWh	DQ status for cumulative active-energy direction.	From `silver.fact_energy_readings_flagged`.	DQ-KWH-001
52	overall_status	text	YES	—	—	No/Not shown	—	Worst status across the 12 status inputs included by the live `worst_status()` call; current statuses are excluded.	From `silver.fact_energy_readings_flagged`.	—

19.22 `silver.fact_energy_readings_staging`

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
1	energy_meter_key	uuid	YES	—	—	No/Not shown	—	Canonical reporting-device / energy-meter identifier used across Silver and Gold.	Inherited from canonical meter mapping or source telemetry packet.	—
2	source_energy_meter_id	uuid	YES	—	—	No/Not shown	—	Source `energy_meters.id` identifier retained for traceability.	Inherited from canonical meter mapping or source telemetry packet.	—
3	facility_key	uuid	YES	—	—	No/Not shown	—	Facility identifier associated with the record.	Inherited from canonical meter mapping or source telemetry packet.	—
4	zone_key	uuid	YES	—	—	No/Not shown	—	Zone/unit identifier associated with the record.	Inherited from canonical meter mapping or source telemetry packet.	—
5	grid_billing_account_key	uuid	YES	—	—	No/Not shown	—	Grid billing-account identifier associated with the record.	Inherited from canonical meter mapping or source telemetry packet.	—
6	date_key	integer	YES	—	—	No/Not shown	—	Calendar date surrogate/key in YYYYMMDD-style integer form.	Inherited from canonical meter mapping or source telemetry packet.	—

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
7	time_key	integer	YES	—	—	No/Not shown	—	Time key derived from the reading timestamp.	Inherited from canonical meter mapping or source telemetry packet.	—
8	source_timestamp_ms	bigint	YES	—	—	No/Not shown	—	Original Bronze telemetry epoch timestamp in milliseconds.	Inherited from canonical meter mapping or source telemetry packet.	—
9	reading_timestamp_utc	timestamp without time zone	YES	—	—	No/Not shown	UTC timestamp	Telemetry reading timestamp in UTC at native source precision.	Converted from Bronze epoch milliseconds to UTC timestamp.	—
10	voltage_line_1_v	double precision	YES	—	—	No/Not shown	V	Electrical measurement pivoted from Bronze telemetry key `V1`.	Pivot from Bronze telemetry key `V1` via Silver landing views.	DQ-VL-001
11	voltage_line_2_v	double precision	YES	—	—	No/Not shown	V	Electrical measurement pivoted from Bronze telemetry key `V2`.	Pivot from Bronze telemetry key `V2` via Silver landing views.	DQ-VL-001
12	voltage_line_3_v	double precision	YES	—	—	No/Not shown	V	Electrical measurement pivoted from Bronze telemetry key `V3`.	Pivot from Bronze telemetry key `V3` via Silver landing views.	DQ-VL-001
13	voltage_line_average_v	double precision	YES	—	—	No/Not shown	V	Electrical measurement pivoted from Bronze	Pivot from Bronze telemetry key `Va` via Silver	DQ-VL-001

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
								telemetry key `Va`.	landing views.	
14	voltage_neutral_1_v	double precision	YES	—	—	No/Not shown	V	Electrical measurement pivoted from Bronze telemetry key `Vn1`.	Pivot from Bronze telemetry key `Vn1` via Silver landing views.	DQ-VN-001
15	voltage_neutral_2_v	double precision	YES	—	—	No/Not shown	V	Electrical measurement pivoted from Bronze telemetry key `Vn2`.	Pivot from Bronze telemetry key `Vn2` via Silver landing views.	DQ-VN-001
16	voltage_neutral_3_v	double precision	YES	—	—	No/Not shown	V	Electrical measurement pivoted from Bronze telemetry key `Vn3`.	Pivot from Bronze telemetry key `Vn3` via Silver landing views.	DQ-VN-001
17	voltage_neutral_average_v	double precision	YES	—	—	No/Not shown	V	Electrical measurement pivoted from Bronze telemetry key `Vna`.	Pivot from Bronze telemetry key `Vna` via Silver landing views.	DQ-VN-001
18	current_phase_1_a	double precision	YES	—	—	No/Not shown	A	Electrical measurement pivoted from Bronze telemetry key `Amp1`.	Pivot from Bronze telemetry key `Amp1` via Silver landing views.	DQ-CUR-001
19	current_phase_2_a	double precision	YES	—	—	No/Not shown	A	Electrical measurement pivoted from Bronze telemetry key `Amp2`.	Pivot from Bronze telemetry key `Amp2` via Silver landing views.	DQ-CUR-001
20	current_phase_3_a	double precision	YES	—	—	No/Not shown	A	Electrical measurement	Pivot from Bronze	DQ-CUR-001

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
								pivoted from Bronze telemetry key `Amp3`.	telemetry key `Amp3` via Silver landing views.	
21	current_average_a	double precision	YES	—	—	No/Not shown	A	Electrical measurement pivoted from Bronze telemetry key `Ampa`.	Pivot from Bronze telemetry key `Ampa` via Silver landing views.	—
22	frequency_hz	double precision	YES	—	—	No/Not shown	Hz	Electrical measurement pivoted from Bronze telemetry key `Hz`.	Pivot from Bronze telemetry key `Hz` via Silver landing views.	DQ-FREQ-001
23	power_factor_phase_1	double precision	YES	—	—	No/Not shown	unitless	Electrical measurement pivoted from Bronze telemetry key `PF1`.	Pivot from Bronze telemetry key `PF1` via Silver landing views.	DQ-PF-001
24	power_factor_phase_2	double precision	YES	—	—	No/Not shown	unitless	Electrical measurement pivoted from Bronze telemetry key `PF2`.	Pivot from Bronze telemetry key `PF2` via Silver landing views.	DQ-PF-001
25	power_factor_phase_3	double precision	YES	—	—	No/Not shown	unitless	Electrical measurement pivoted from Bronze telemetry key `PF3`.	Pivot from Bronze telemetry key `PF3` via Silver landing views.	DQ-PF-001
26	power_factor_average	double precision	YES	—	—	No/Not shown	unitless	Electrical measurement pivoted from Bronze telemetry key `PFa`.	Pivot from Bronze telemetry key `PFa` via Silver landing views.	DQ-PF-001

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
27	active_power_phase_1_kw	double precision	YES	—	—	No/Not shown	kW	Electrical measurement pivoted from Bronze telemetry key `kW1`.	Pivot from Bronze telemetry key `kW1` via Silver landing views.	—
28	active_power_phase_2_kw	double precision	YES	—	—	No/Not shown	kW	Electrical measurement pivoted from Bronze telemetry key `kW2`.	Pivot from Bronze telemetry key `kW2` via Silver landing views.	—
29	active_power_phase_3_kw	double precision	YES	—	—	No/Not shown	kW	Electrical measurement pivoted from Bronze telemetry key `kW3`.	Pivot from Bronze telemetry key `kW3` via Silver landing views.	—
30	active_power_total_kw	double precision	YES	—	—	No/Not shown	kW	Electrical measurement pivoted from Bronze telemetry key `TkW`.	Pivot from Bronze telemetry key `TkW` via Silver landing views.	—
31	apparent_power_phase_1_kva	double precision	YES	—	—	No/Not shown	kVA	Electrical measurement pivoted from Bronze telemetry key `kVA1`.	Pivot from Bronze telemetry key `kVA1` via Silver landing views.	—
32	apparent_power_phase_2_kva	double precision	YES	—	—	No/Not shown	kVA	Electrical measurement pivoted from Bronze telemetry key `kVA2`.	Pivot from Bronze telemetry key `kVA2` via Silver landing views.	—
33	apparent_power_phase_3_kva	double precision	YES	—	—	No/Not shown	kVA	Electrical measurement pivoted from Bronze	Pivot from Bronze telemetry key `kVA3`	—

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
								telemetry key `kVA3`.	via Silver landing views.	
34	apparent_power_total_kva	double precision	YES	—	—	No/Not shown	kVA	Electrical measurement pivoted from Bronze telemetry key `TkVA`.	Pivot from Bronze telemetry key `TkVA` via Silver landing views.	—
35	reactive_power_phase_1_kvar	double precision	YES	—	—	No/Not shown	kVAr	Electrical measurement pivoted from Bronze telemetry key `kVAr1`.	Pivot from Bronze telemetry key `kVAr1` via Silver landing views.	—
36	reactive_power_phase_2_kvar	double precision	YES	—	—	No/Not shown	kVAr	Electrical measurement pivoted from Bronze telemetry key `kVAr2`.	Pivot from Bronze telemetry key `kVAr2` via Silver landing views.	—
37	reactive_power_phase_3_kvar	double precision	YES	—	—	No/Not shown	kVAr	Electrical measurement pivoted from Bronze telemetry key `kVAr3`.	Pivot from Bronze telemetry key `kVAr3` via Silver landing views.	—
38	reactive_power_total_kvar	double precision	YES	—	—	No/Not shown	kVAr	Electrical measurement pivoted from Bronze telemetry key `TkVAr`.	Pivot from Bronze telemetry key `TkVAr` via Silver landing views.	—
39	active_energy_cumulative_kwh	double precision	YES	—	—	No/Not shown	kWh	Cumulative active-energy register reported by the meter.	Pivot from Bronze telemetry key `kWh` via Silver landing views.	DQ-KWH-001

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
40	apparent_energy_cumulative_kvah	double precision	YES	—	—	No/Not shown	kVAh	Electrical measurement pivoted from Bronze telemetry key `kVAh`.	Pivot from Bronze telemetry key `kVAh` via Silver landing views.	—
41	reactive_energy_cumulative_kvarh	double precision	YES	—	—	No/Not shown	kVArh	Electrical measurement pivoted from Bronze telemetry key `kVArh`.	Pivot from Bronze telemetry key `kVArh` via Silver landing views.	—
42	source_measurement_row_count	integer	YES	—	—	No/Not shown	—	Count of source telemetry rows contributing to the meter×timestamp packet.	Derived by packet-level row/distinct-key counts.	DQ-PACKET-001
43	populated_measurement_count	integer	YES	—	—	No/Not shown	—	Count of distinct telemetry keys populated in the meter×timestamp packet.	Derived by packet-level row/distinct-key counts.	DQ-PACKET-001
44	duplicate_measurement_row_count	integer	YES	—	—	No/Not shown	—	Source telemetry rows beyond the distinct-key count for the packet; diagnostic for duplicate measurements.	Derived by packet-level row/distinct-key counts.	DQ-PACKET-001
45	reading_timestamp_saudi	timestamp without time zone	YES	—	—	No/Not shown	Asia/Riyadh timestamp	Telemetry reading timestamp converted to Asia/	Derived from source epoch timestamp using	—

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
								Riyadh local time.	`Asia/Riyadh`.	
46	reading_date_saudi	date	YES	—	—	No/Not shown	—	Saudi-local calendar date derived from the telemetry timestamp.	Derived from source epoch timestamp using `Asia/Riyadh`.	—
47	reading_time_saudi	time without time zone	YES	—	—	No/Not shown	—	Reading time saudi.	Derived from source epoch timestamp using `Asia/Riyadh`.	—
48	saudi_date_key	integer	YES	—	—	No/Not shown	—	Saudi-local date key derived from reading timestamp.	Derived from source epoch timestamp using `Asia/Riyadh`.	—
49	saudi_time_key	integer	YES	—	—	No/Not shown	—	Saudi-local time key derived from reading timestamp.	Derived from source epoch timestamp using `Asia/Riyadh`.	—
50	calculated_current_average_a	double precision	YES	—	—	No/Not shown	A	Calculated arithmetic mean of the three phase-current readings when evaluation prerequisites are met.	Derived in `fact_energy_readings_staging` from three-phase values.	DQ-IMB-CUR-001
51	current_phase_spread_a	double precision	YES	—	—	No/Not shown	A	Difference between maximum and minimum	Derived in `fact_energy_readings_st`	DQ-CUR-001

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
								three-phase current.	`aging` from three-phase values.	
52	current_imbalance_pct	double precision	YES	—	—	No/Not shown	%	Maximum phase deviation from the three-phase mean, expressed as a percentage of the absolute mean.	Derived in `fact_energy_readings_staging` from three-phase values.	DQ-IMB-CUR-001
53	current_imbalance_evaluation_status	text	YES	—	—	No/Not shown	—	Diagnostic status indicating whether current imbalance was evaluated and why it may be not applicable.	Derived in `fact_energy_readings_staging` from three-phase values.	DQ-IMB-CUR-001
54	calculated_voltage_average_v	double precision	YES	—	—	No/Not shown	V	Calculated arithmetic mean of the three line-voltage phase readings when prerequisites are met.	Derived in `fact_energy_readings_staging` from three-phase values.	DQ-IMB-VOLT-001
55	voltage_phase_spread_v	double precision	YES	—	—	No/Not shown	V	Difference between maximum and minimum three-phase line voltage.	Derived in `fact_energy_readings_staging` from three-phase values.	DQ-IMB-VOLT-001

Ord	Column	Type	Nullable	Default	Key	Indexed	Unit / Format	Business definition	Source / transformation	DQ rule
56	voltage_imbalance_pct	double precision	YES	—	—	No/Not shown	%	Maximum phase-voltage deviation from the three-phase mean, expressed as a percentage of the absolute mean.	Derived in `fact_energy_readings_staging` from three-phase values.	DQ-IMB-VOLT-001
57	voltage_imbalance_evaluation_status	text	YES	—	—	No/Not shown	—	Diagnostic status indicating whether voltage imbalance was evaluated and why it may be not applicable.	Derived in `fact_energy_readings_staging` from three-phase values.	DQ-IMB-VOLT-001

20. Metadata Evidence & Change Control

20.1 Evidence used for this edition

This edition incorporates live PostgreSQL outputs for: - object type and relation size; - complete scoped column metadata from `information_schema.columns`; - PK/FK/CHECK/NOT NULL metadata from PostgreSQL catalogs; - index definitions from `pg_indexes`; - live view definitions via `pg_get_viewdef()`; - live function definitions via `pg_get_functiondef()`.

20.2 Important implementation confirmations from the live metadata

1. `silver.fact_energy_readings_flagged` calls **`classify_current_v2(value, cb_max, phase_kw)`**, not the two-argument overload.
2. Current/power mismatch logic is confined to the **lower fringe (`current < 15% of CB`)** in that production function.
3. Current uses literal zero comparisons; no epsilon is present in the extracted function.
4. Negative current returns `Invalid`; current strictly above breaker rating returns `Critical`.
5. `overall_status` excludes **all three current statuses**.
6. `silver.fact_energy_readings_snapshot` is a physical table with enforced PK `(energy_meter_key, reading_timestamp_utc)`.

7. `silver.fact_energy_consumption_validated` is a physical table with indexes on Saudi date, meter label, and `(energy_meter_key, reading_timestamp_utc)`, but no PK/UNIQUE constraint was returned.
8. `gold.energy_reading_gaps` is a physical table; its scheduler/refresh procedure is not present in the supplied catalog metadata.
9. `gold.energy_raw` is a SQL view with 52 physical columns. `meter_label` is not one of them; the readable meter label used in Power BI is a model-level calculated/related field as documented earlier.

20.3 Change-control principle

Future editions should re-run the metadata extractor after schema/function changes. Physical metadata should be replaced from the new catalog output rather than manually guessed. Business definitions and DQ descriptions should then be reviewed only where the underlying SQL or approved business rule changed.